Key words and Terms

thermodynamic cycle	热力学循环
heat engine	热机
efficiency of heat engine	热机效率
refrigerator	制冷机
heat pump	热泵
coefficient of performance	制冷系数
reversible (irreversible) process	可逆（不可逆）过程
entropy	熵

Chapter 15   Second Law of Thermodynamics

15.1 Cycle

15.2 Second Law of Thermodynamics

15.3 Entropy

15.1 Cycle

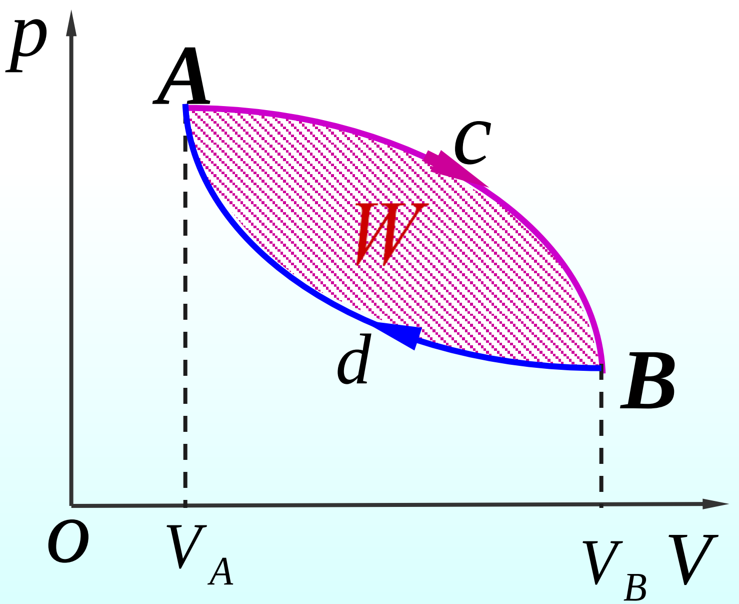
1. Cycle

The system returns repeatedly to its starting point.
若系统经历了一系列准静态过程之后又回到初态，
则称这个系统经历了一个热力学循环过程，简称循环

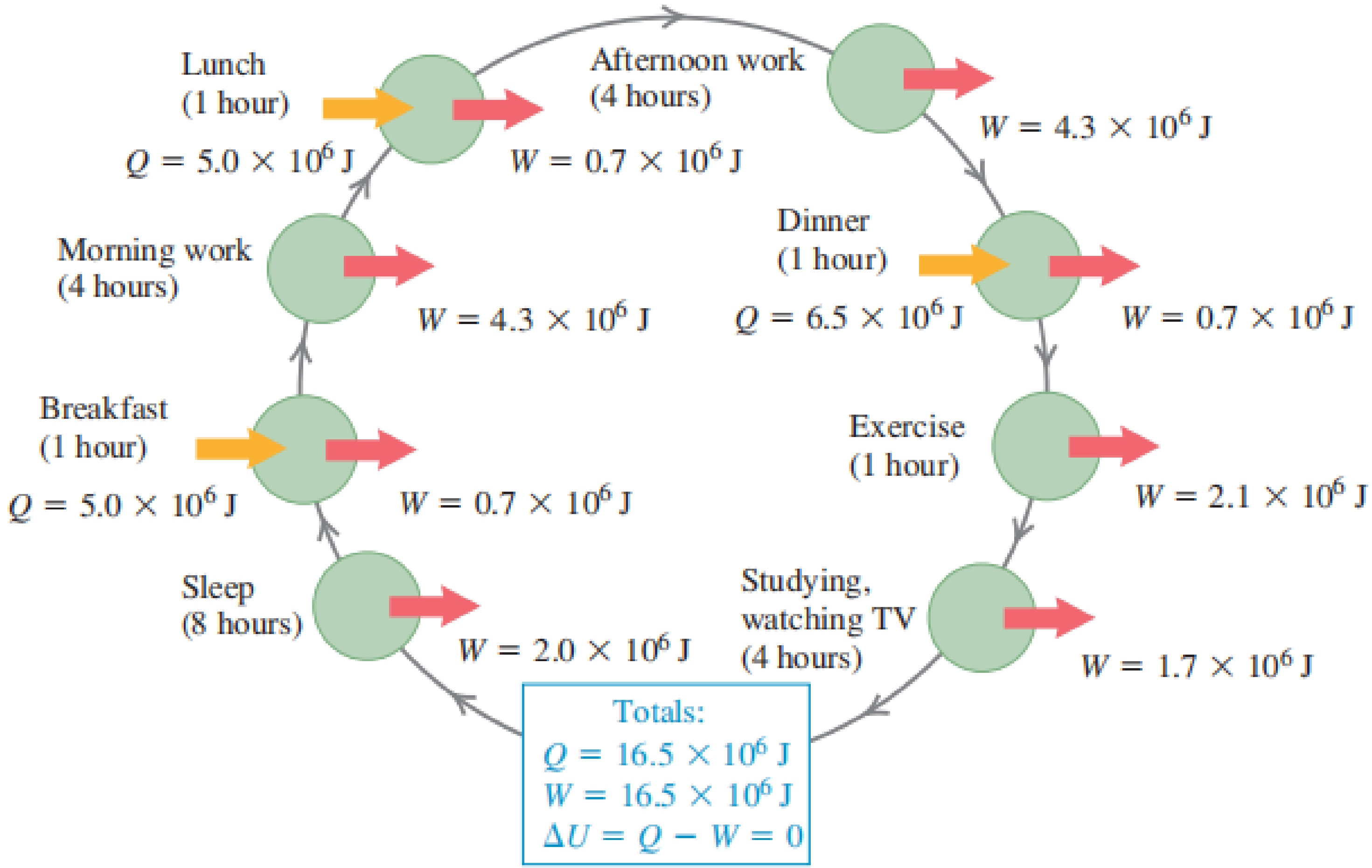
Feature:

$$\Delta U = 0$$

The net work done in a cyclic process is the area enclosed by the curve representing the process on a *PV* diagram.



Every day, your body (a thermodynamic system) goes through a cyclic thermodynamic process like this one. Heat Q is added by metabolizing food, and your body does work W in breathing, walking, and other activities. If you return to the same state at the end of the day, and the net change in your internal energy is zero.

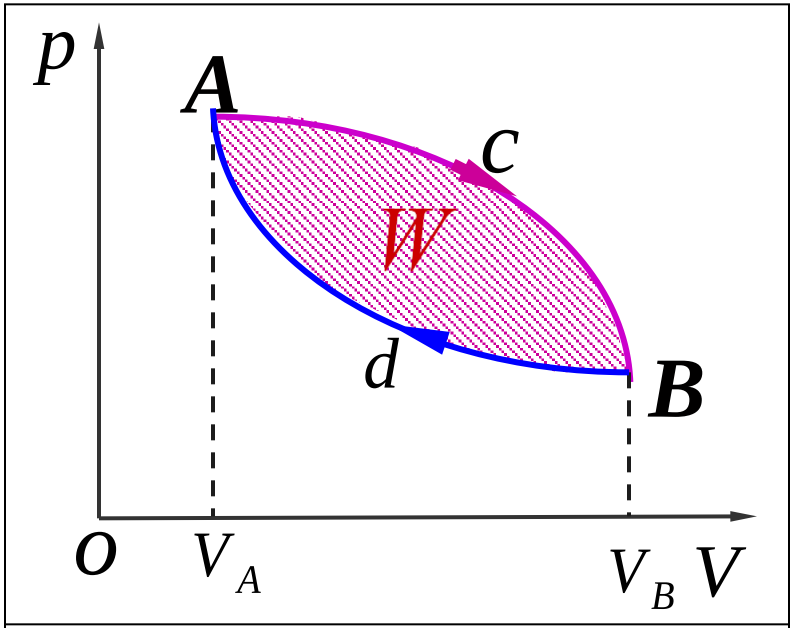


Categories of cycle

(1) Clockwise cycle(正循环):

***W* >0**

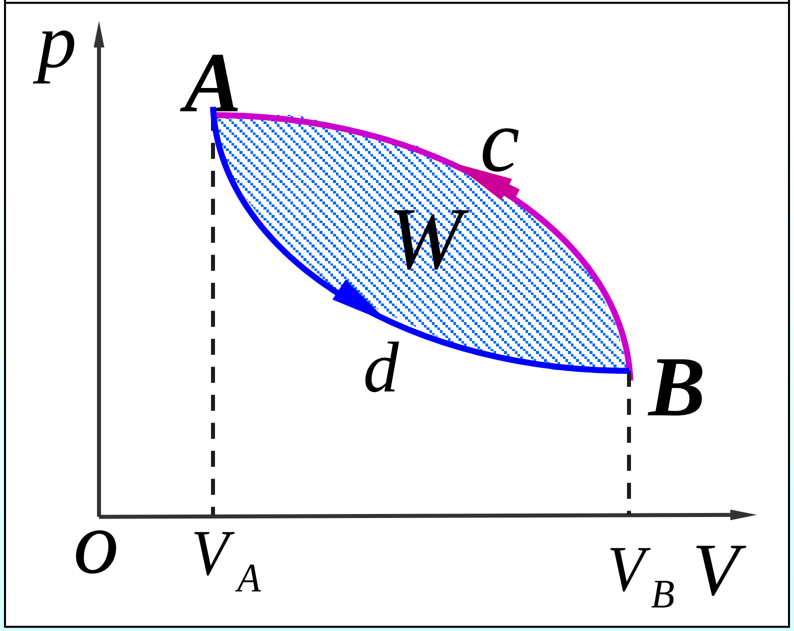
Heat engine cycle



(2) Anticlockwise cycle (逆循环):

***W* <0**

Cooling device cycle



2. Heat engine efficiency and coefficient of performance

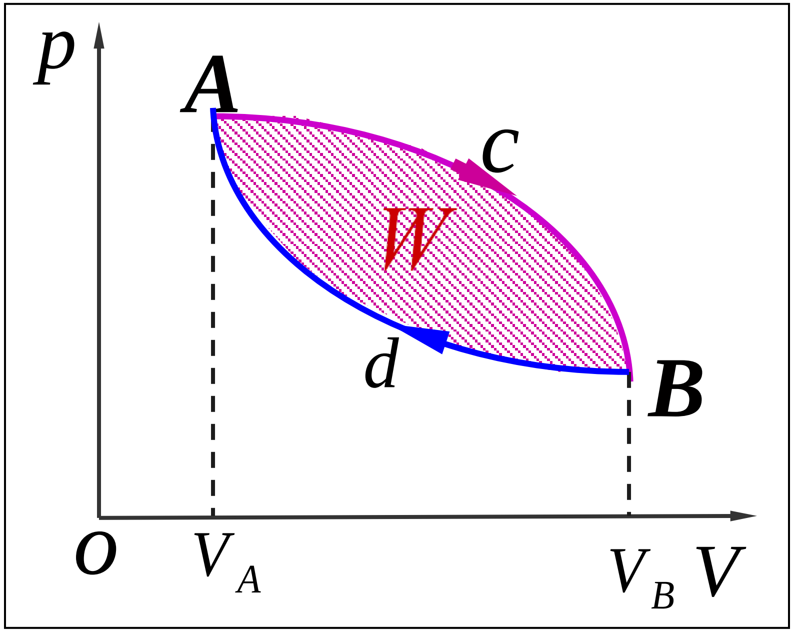
Heat Engine（热机）

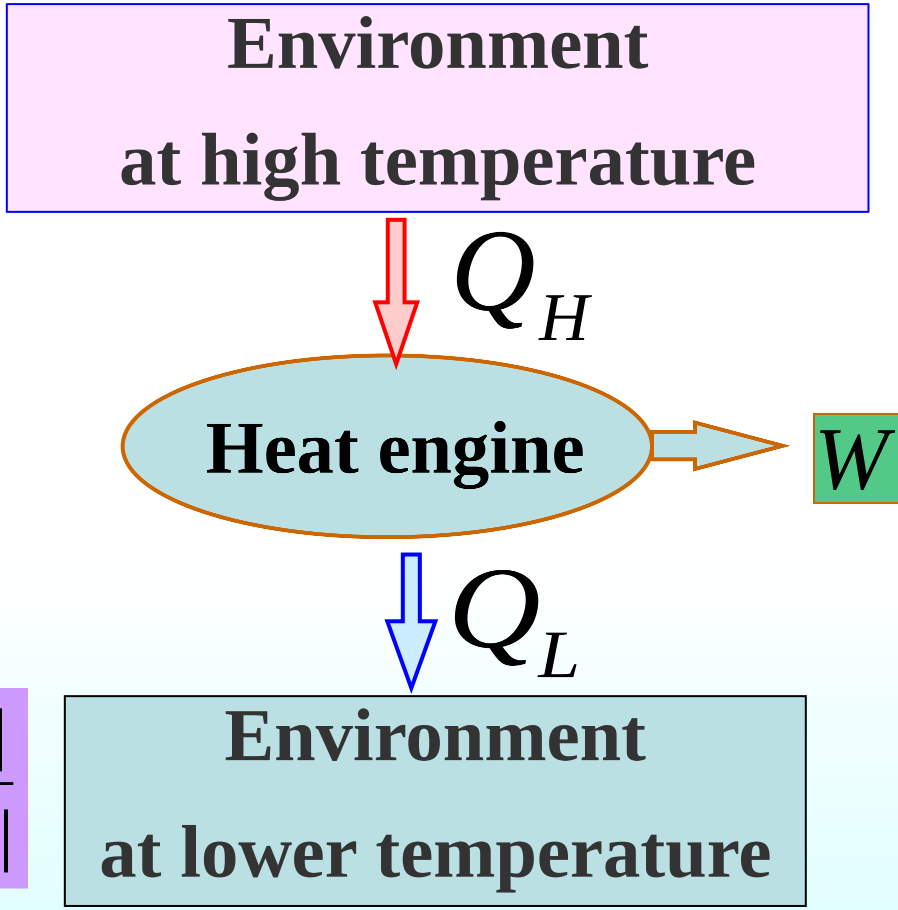
A heat engine is any device that changes energy into mechanical work, such as steam engines and automobile Engines.

Cooling device (制冷机）

A cooling device is to transform heat out of a cool environment into a warm environment.

(1) Heat engine efficiency *e*



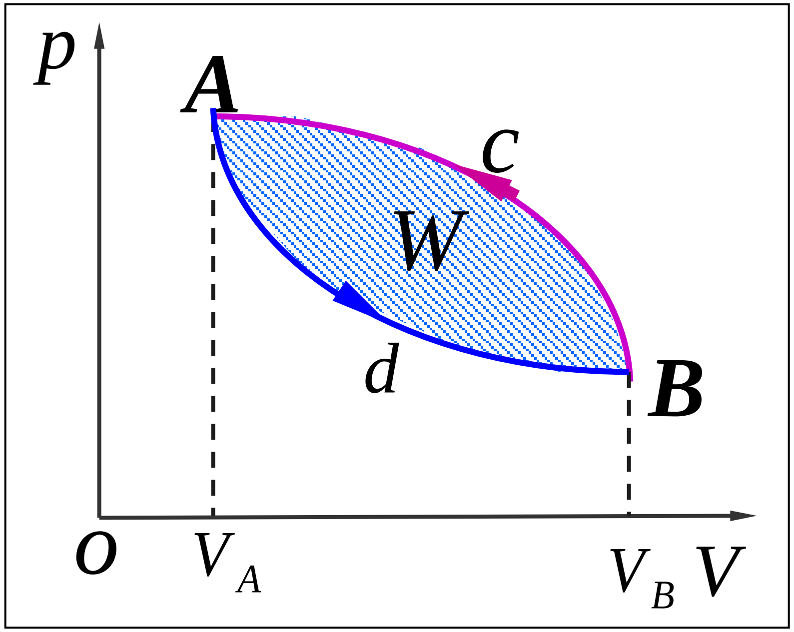


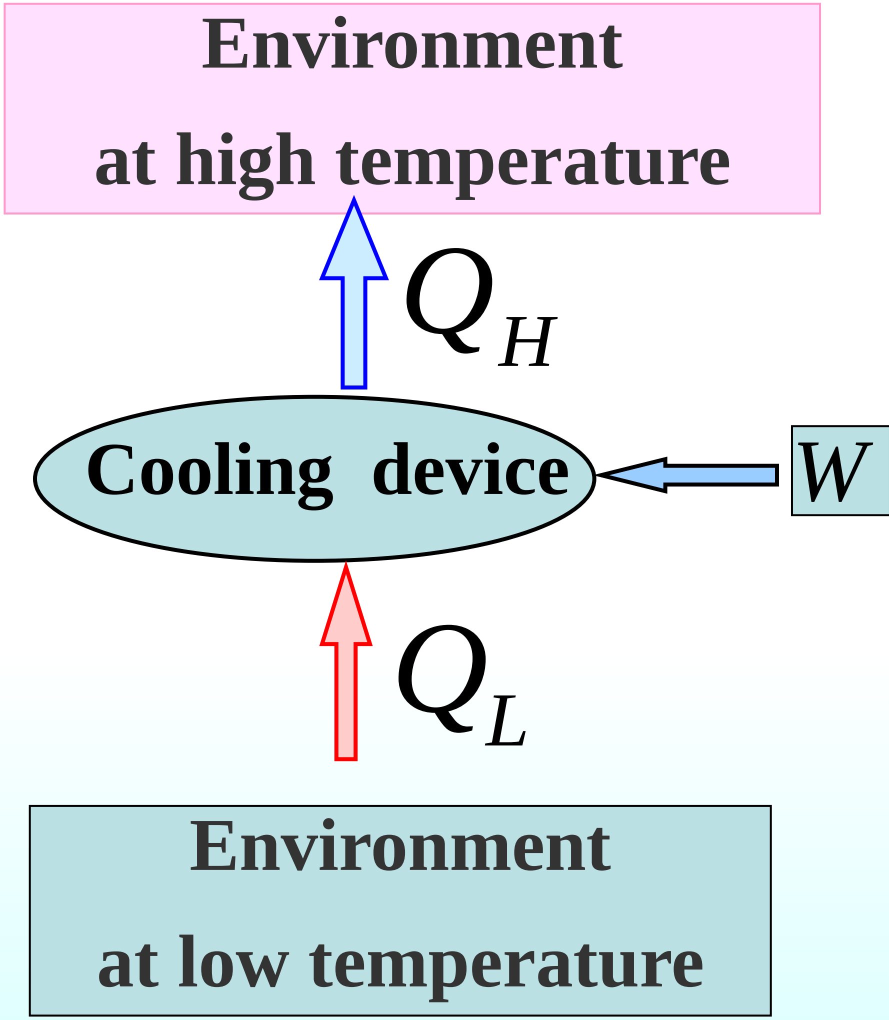
Heat engine efficiency:

$$e=\frac{|W|}{|Q_H|}=\frac{|Q_H|-|Q_L|}{|Q_H|}=1-\frac{|Q_L|}{|Q_H|}$$

$$e=\frac{\text{energy we get (mechanical work)}}{\text{energy we pay for (energy transfer at high temperature)}}$$

(2) Coefficient of Performance *CP*





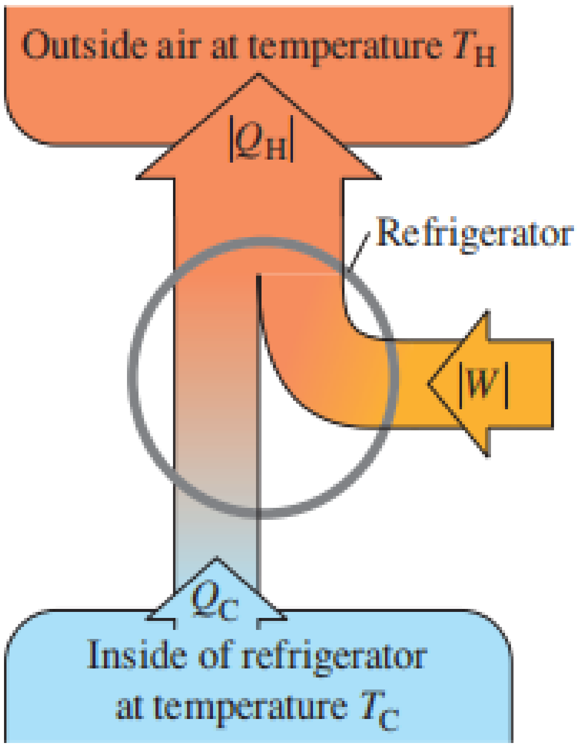
Coefficient of Performance:

$$CP(K) = \frac{\text{what we want}}{\text{what we pay for}}$$

The effectiveness of a refrigerator

$$CP(K)[\text{cooling mode}] = \frac{\text{Energy removed from the cold reservoir}}{\text{Work done by refrigerator}}$$

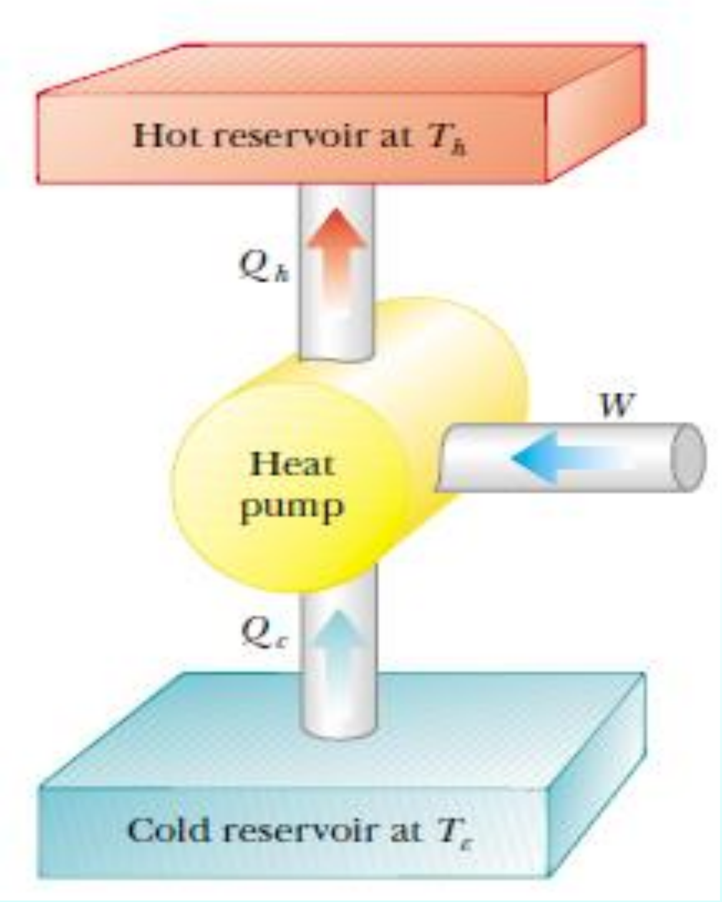
$$\boxed{CP = \frac{|Q_L|}{|W|} = \frac{|Q_L|}{|Q_H| - |Q_L|}}$$



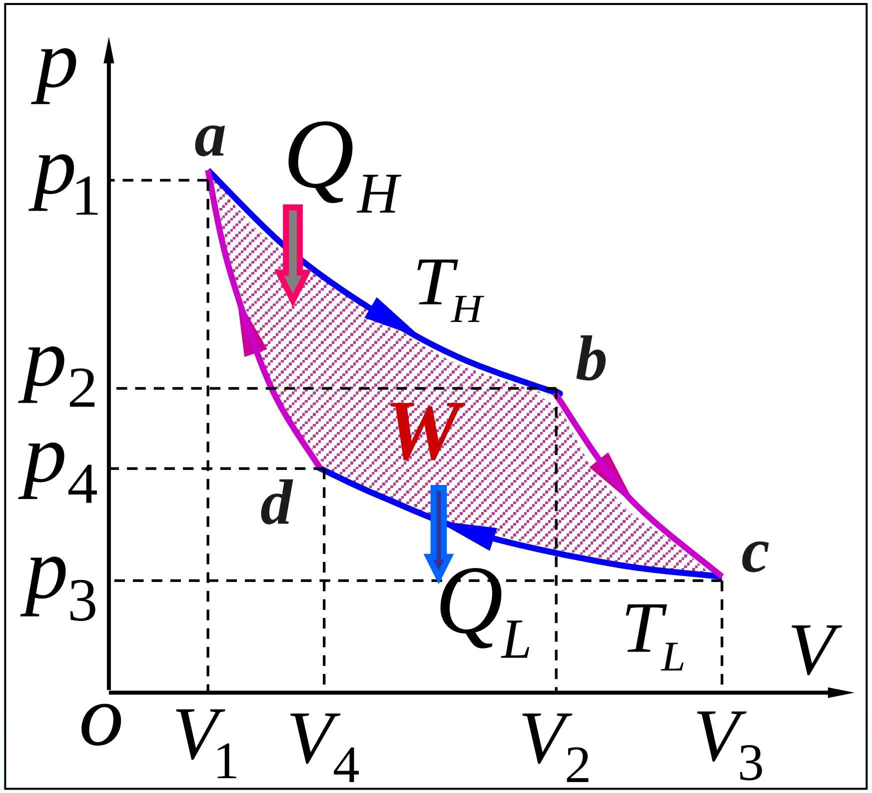
The effectiveness of a heat pump

$$CP(K)[\text{heating mode}] = \frac{\text{Energy transferred at high temperature}}{\text{Work done by pump}}$$

$$CP = \frac{|Q_H|}{|W|} = \frac{|Q_H|}{|Q_H| - |Q_L|}$$



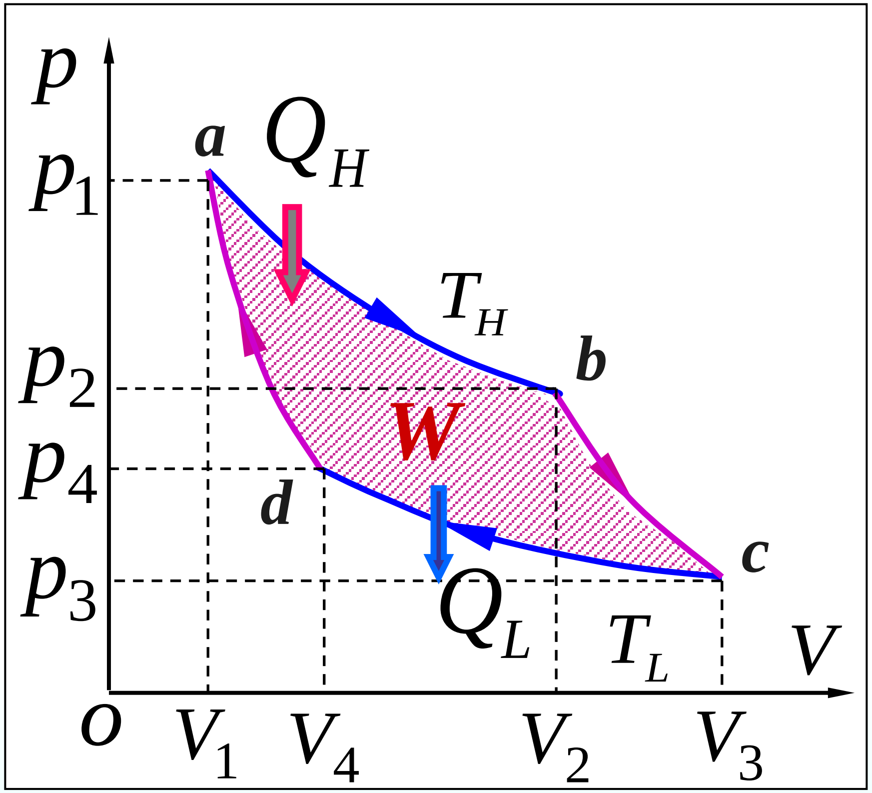
3. Carnot Cycle

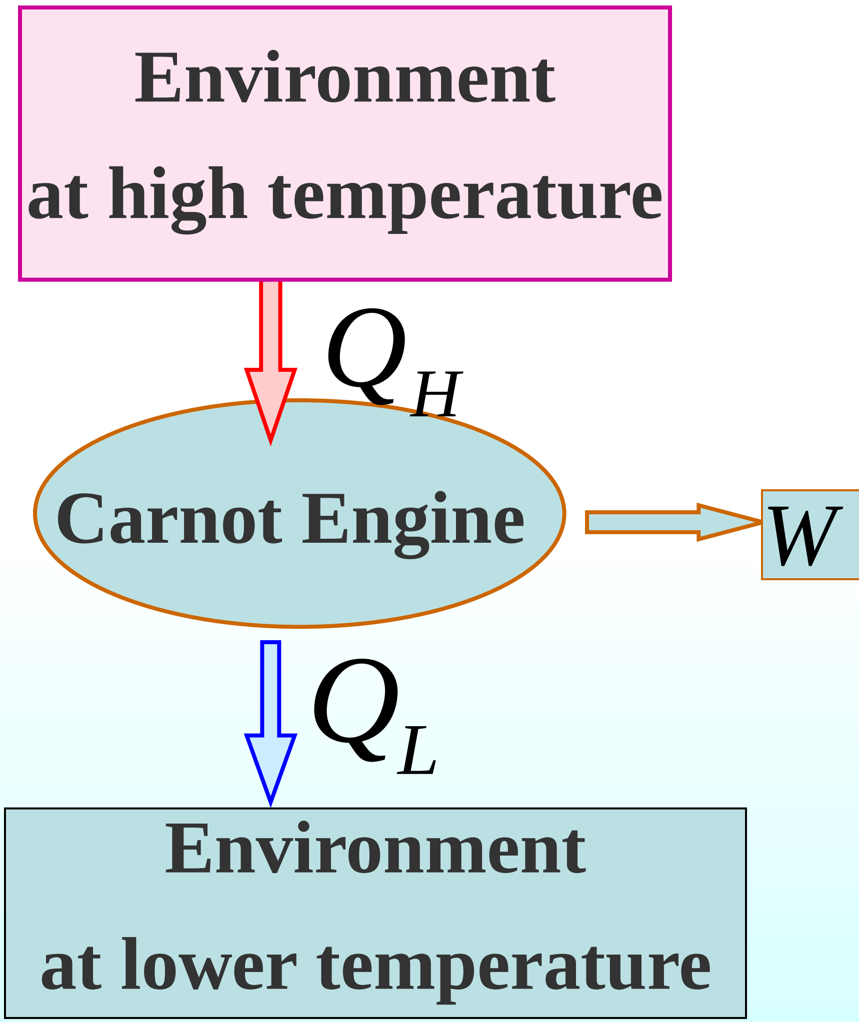


French engineer， Sadi Carnot (1796–1832).

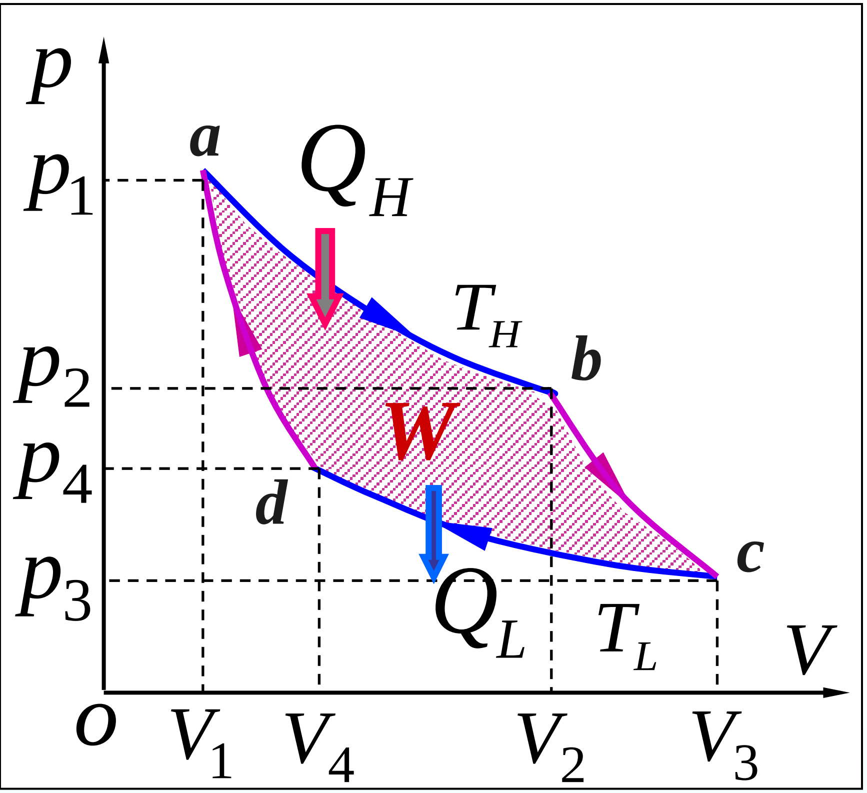
A Carnot cycle consists of two isothermal and two adiabatic processes.

(1) Carnot heat engine efficiency 卡诺热机效率





Carnot Cycle



a-b Isothermal expansion $Q > 0$

$$|Q_H| = nRT_H \ln\frac{V_2}{V_1}$$

b-c Adiabatic expansion $W > 0$

c-d Isothermal compression $Q < 0$

$$|Q_L| = \left| nRT_L \ln\frac{V_4}{V_3} \right| = nRT_L \ln\frac{V_3}{V_4}$$

d-a Adiabatic compression $W < 0$

b-c, d-a are adiabatic processes:

$$V_2^{\gamma-1}T_H = V_3^{\gamma-1}T_L \quad V_1^{\gamma-1}T_H = V_4^{\gamma-1}T_L \Rightarrow \frac{V_2}{V_1} = \frac{V_3}{V_4}$$

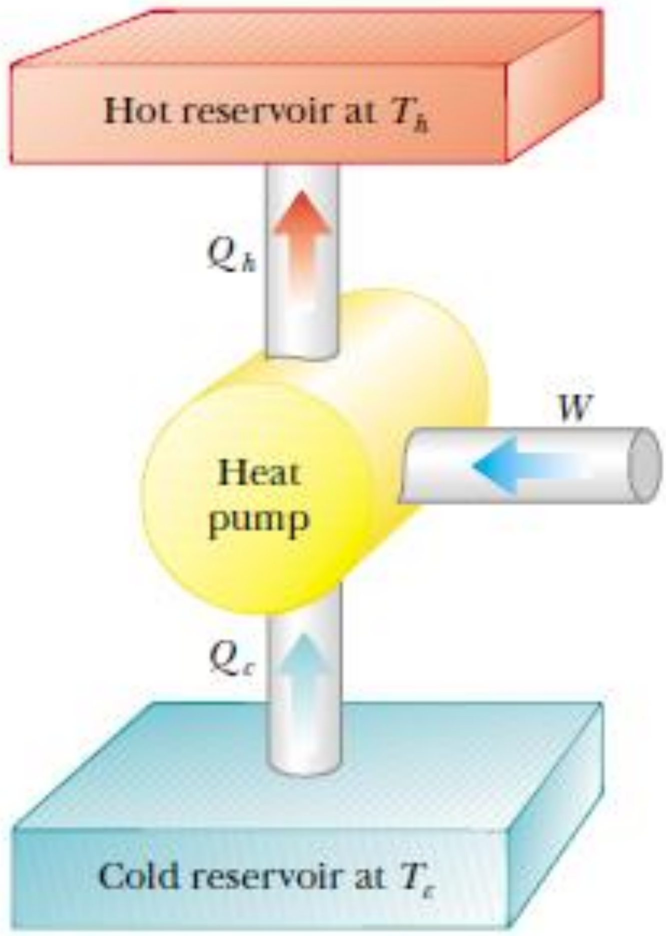
The efficiency of a Carnot cycle:

$$e = 1 - \frac{|Q_L|}{|Q_H|} = 1 - \frac{T_L}{T_H}\frac{\ln\frac{V_3}{V_4}}{\ln\frac{V_2}{V_1}} = 1 - \frac{T_L}{T_H}$$

The efficiency of a Carnot engine depends **only** on the temperatures T_H and T_L. The efficiency is large when the **temperature difference** is large, and it is very small when the temperatures are nearly equal.

卡诺热机效率与工作物质无关，只与两个热源的温度有关，两热源的温差越大，则卡诺循环的效率越高 .

(2)*CP* of a Carnot cycle (heat pump or refrigerator)



$$CP(\text{cooling mode}) = \frac{|Q_L|}{|W|} = \frac{|Q_L|}{|Q_H| - |Q_L|}$$

$$CP = \frac{T_L}{T_H - T_L}$$

$$CP(\text{heating mode}) = \frac{|Q_H|}{|W|} = \frac{|Q_H|}{|Q_H| - |Q_L|}$$

$$CP = \frac{T_H}{T_H - T_L}$$

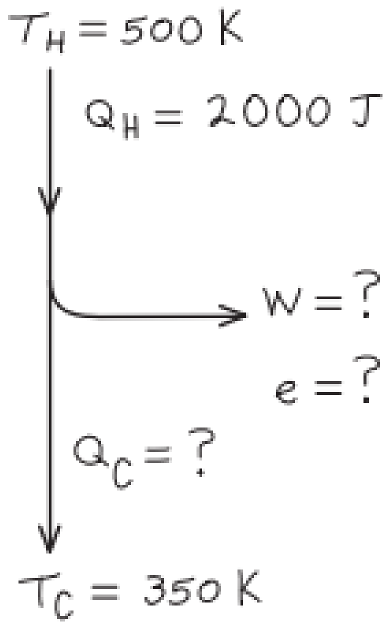
Question :**A Carnot engine takes 2000 J of heat from a reservoir (热源) at 500 K, does some work, and discards some heat to a reservoir at 350 K. How much heat is discarded?**

A. 600J

B. -600J

C. 1400J

D. -1400J



Question :

A steam engine operates between 500^{o}C and 270^{o}C. What is the maximum possible efficiency of this engine?

Solution:

$$e_{ideal} = 1 - \frac{T_L}{T_H} = 1 - \frac{270+273}{500+273} = 30\%$$

Example: A heat pump has a coefficient of performance of 3.0 and is rated to do work at 1500W. (a) How much heat can it add to a room per second? (b) If the heat pump were turned around to act as an air conditioner in the summer, what would you expect its coefficient of performance to be, assuming all else stays the same?

Solution:

$$(a)\left|Q_H\right| = CP \times \left|W\right| = 3.0 \times 1500 = 4500\,J$$

$$(b)\left|Q_L\right| = \left|Q_H\right| - \left|W\right| = 4500 - 1500 = 3000\,J$$

$$CP = \frac{\left|Q_L\right|}{\left|W\right|} = \frac{3000}{1500} = 2.0$$

Example: **An ideal engine has an efficiency of 35 percent. If it were run backward as a heat pump, what would be its coefficient of performance?**

Solution:

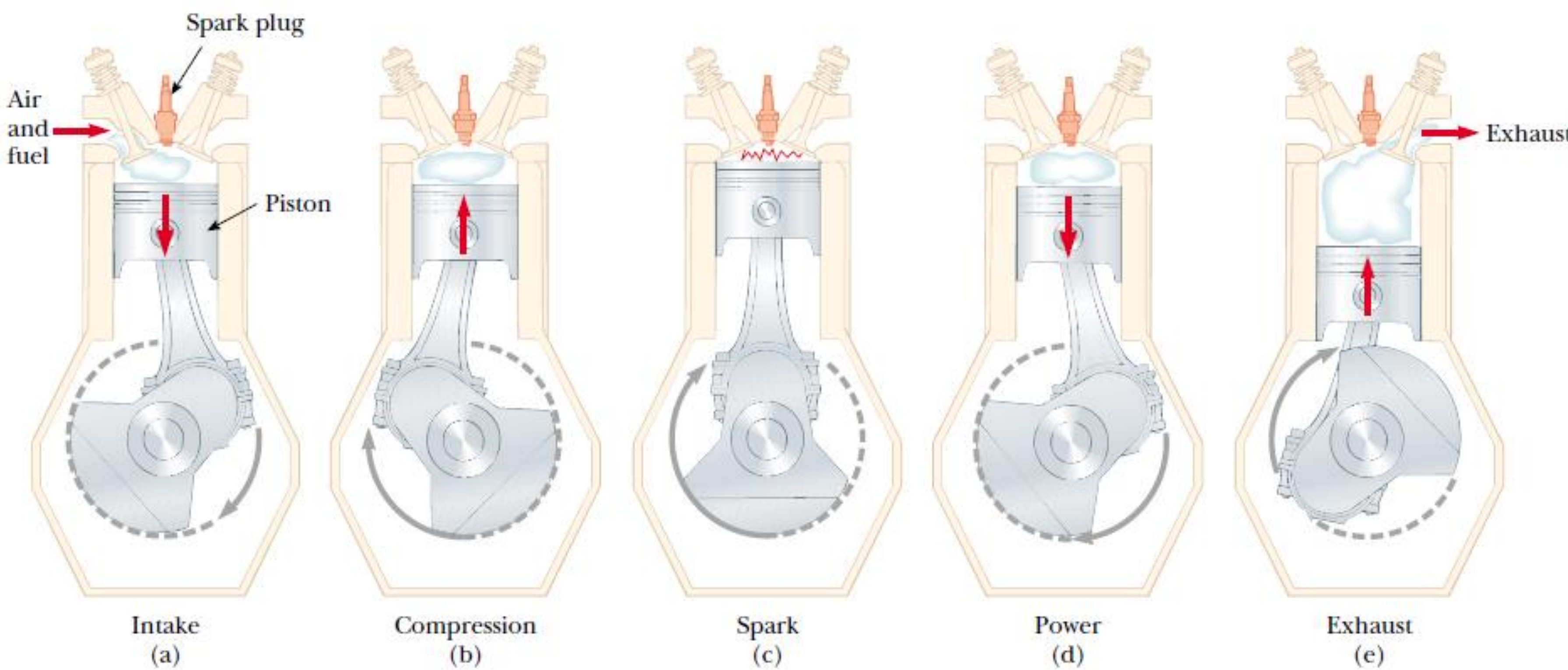
$$e = \frac{|W|}{|Q_H|} = 1 - \frac{|Q_L|}{|Q_H|} = 35\%$$

$$CP = \frac{|Q_H|}{|W|} = \frac{1}{0.35} = 2.86$$

Cycle

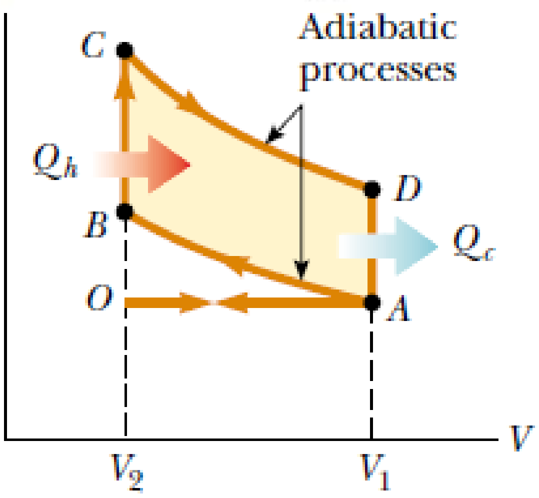
cycle	***W***	***Definiton***	**Carnot Cycle**	***e* or CP**
Heat engine 热机	**>0**	$e=\frac{\lvert W\rvert}{\lvert Q_H\rvert}$	$e=1-\frac{T_L}{T_H}$	e
refrigerator 冰箱模式	**<0**	$CP_{cooling}=\frac{\lvert Q_L\rvert}{\lvert W\rvert}$	$CP_{\text{cooling}}=\frac{T_L}{T_H-T_L}$	$CP_{\text{cooling}}=\frac{1}{e}-1$
heat pump 热泵模式	**<0**	$CP_{\text{heating}}=\frac{\lvert Q_H\rvert}{\lvert W\rvert}$	$CP_{\text{heating}}=\frac{T_H}{T_H-T_L}$	$CP_{\text{heating}}=\frac{1}{e}$

Internal combustion engine： The Otto Cycle (奥拓循环) .

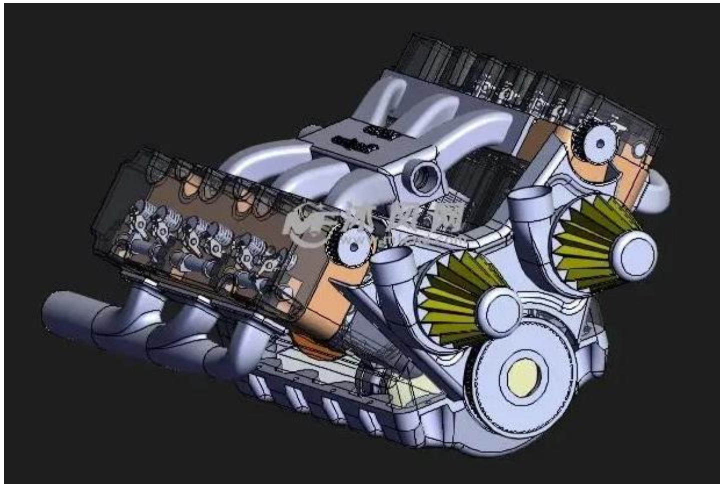


$r = V_1/V_2$ compression ratio

The efficiency increases as the compression ratio increases.



The Otto Cycle. **Paths AD and BC are adiabatic, and paths bd and da are at constant volume.**



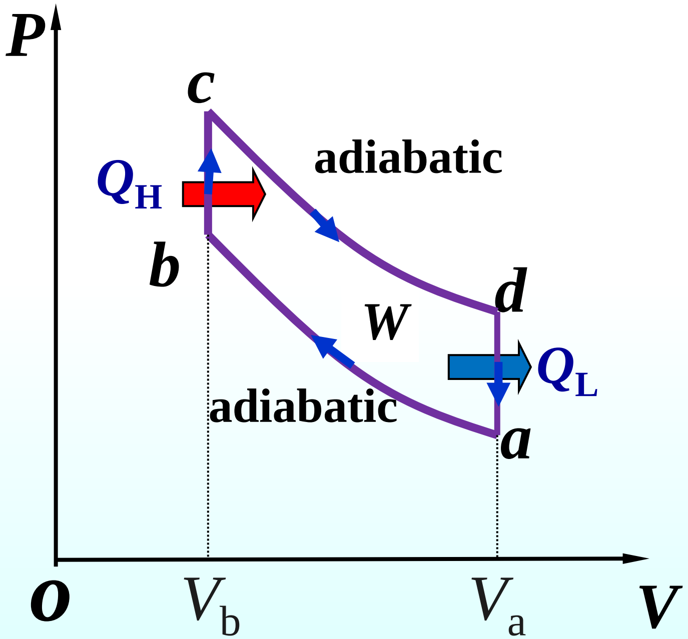
Find: e=?

Solution:

$$e = 1 - \frac{|Q_{da}|}{|Q_{bc}|}$$

$$= 1 - \frac{nC_v(T_d - T_a)}{nC_v(T_c - T_b)}$$

$$= 1 - \frac{T_d - T_a}{T_c - T_b}$$



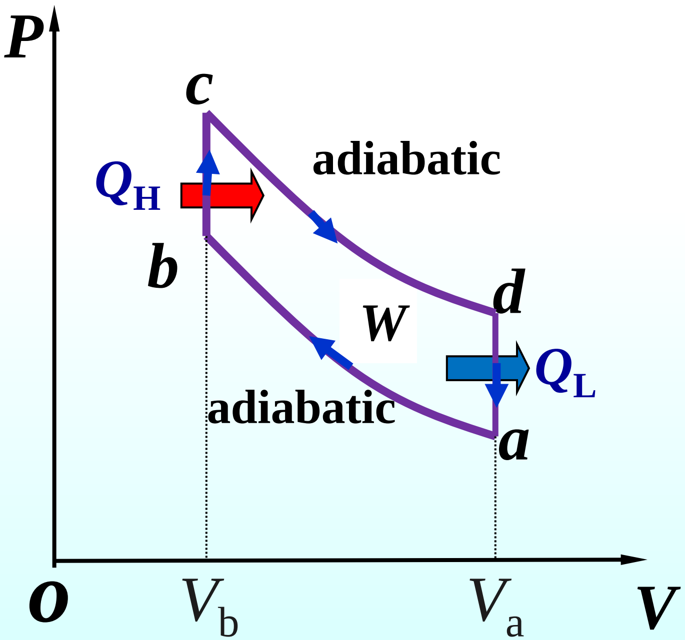
ab and cd are adiabatic processes

$$\frac{T_a}{T_b} = \left(\frac{V_b}{V_a}\right)^{\gamma-1}, \qquad \frac{T_d}{T_c} = \left(\frac{V_c}{V_d}\right)^{\gamma-1}$$

So $$\frac{T_a}{T_b} = \frac{T_d}{T_c}$$

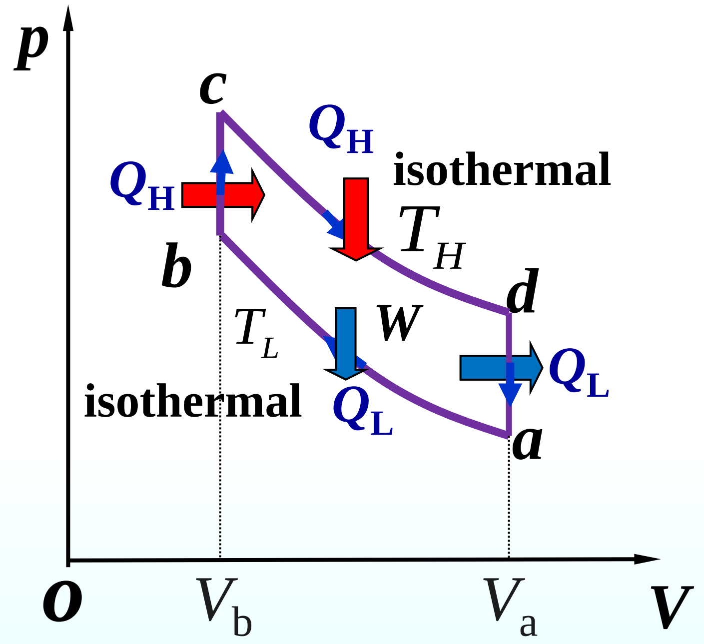
$$e = 1 - \frac{T_d - T_a}{T_c - T_b} = 1 - \frac{T_a}{T_b} = 1 - \frac{T_d}{T_c}$$

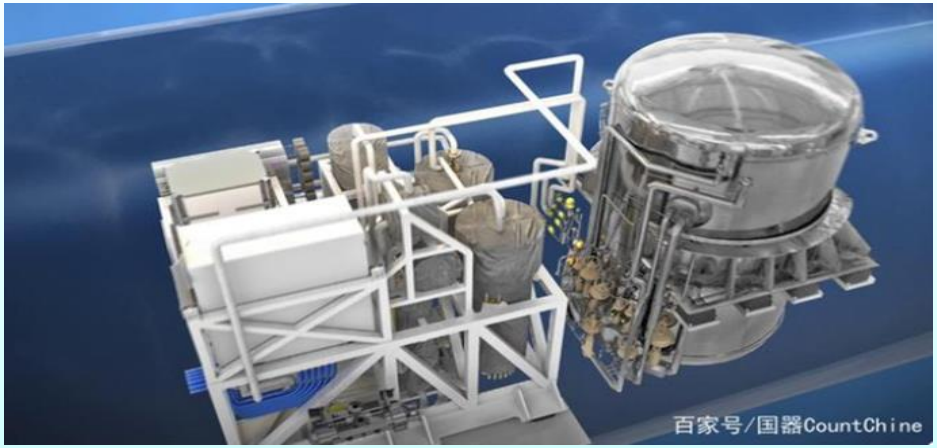
$$= 1 - \left(\frac{V_b}{V_a}\right)^{\gamma-1} = 1 - \frac{1}{r^{\gamma-1}}$$



Stirling cycle (斯特林循环). It is useful to describe external combustion engines as well as solar-power systems. Paths ab and cd are isothermal , and paths bc and da are at constant volume. Find *e* =?

$$e = 1 - \frac{\frac{3}{2}(T_H - T_L) + T_L \ln\frac{V_b}{V_a}}{\frac{3}{2}(T_H - T_L) + T_H \ln\frac{V_b}{V_a}}$$

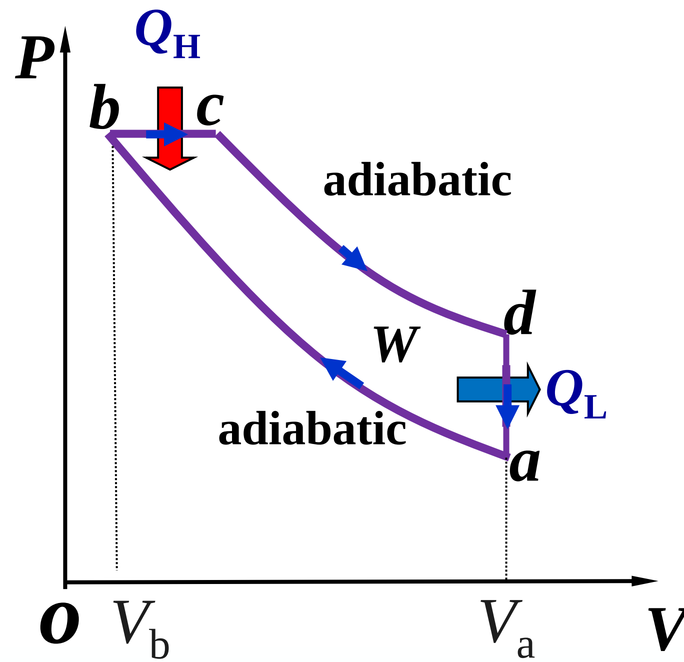




斯特林循环作为外燃机，
其理论效率高。

The Diesel Cycle (迪赛尔循环). **Paths ab and cd are adiabatic. Path bc is isobaric and da is at constant volume. Find *e* =?**

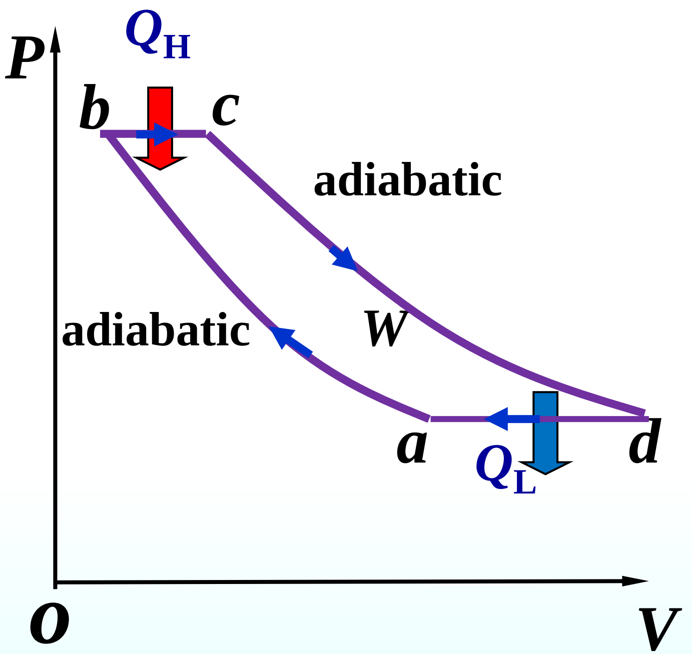
$$e = 1 - \frac{T_d - T_a}{\gamma (T_c - T_b)} = 1 - \frac{[(V_a/V_c)^{\gamma-1} - (V_a/V_b)^{\gamma-1}]}{\gamma[(V_a/V_c)^{-1} - (V_a/V_b)^{-1}]}$$



The Diesel engine is similar in operation to the gasoline engine. The most important difference is that there is no fuel at the beginning of the compression stroke. Because of the high temperature developed during the adiabatic compression, the fuel ignites spontaneously as it is injected; no spark plugs are needed. (常用于柴油机，压燃式）

Brayton cycle (勃朗登循环) . **A heat engine takes cycle as shown in the Fig. Paths ad and bc are adiabatic , and paths ab and cd are isobaric .**

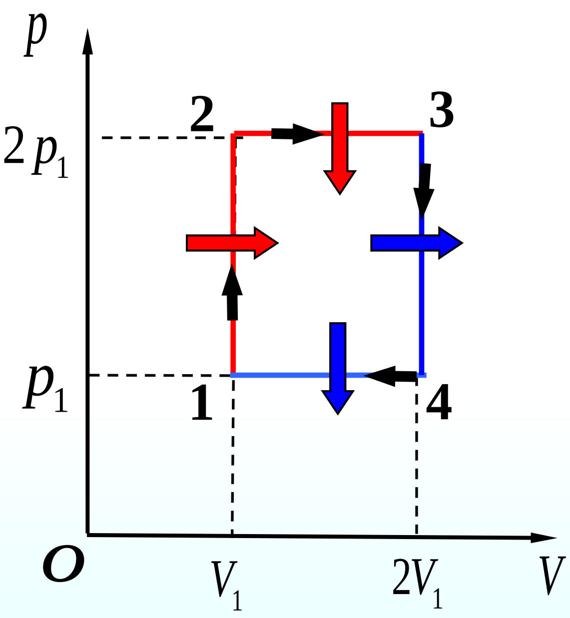
$$e = 1 - \frac{T_d - T_a}{T_c - T_b} = 1 - \frac{T_a}{T_b} = 1 - \left(\frac{P_b}{P_a}\right)^{\frac{1-\gamma}{\gamma}}$$



主要用于燃气轮机

Example: **An ideal diatomic gas of volume V_1 at a pleasure of p_1 varies along the curved path shown in the figure . If p_2=2 P_1, V_2=2$V_{1,}$ find thermal efficiency e=?**

Solution:



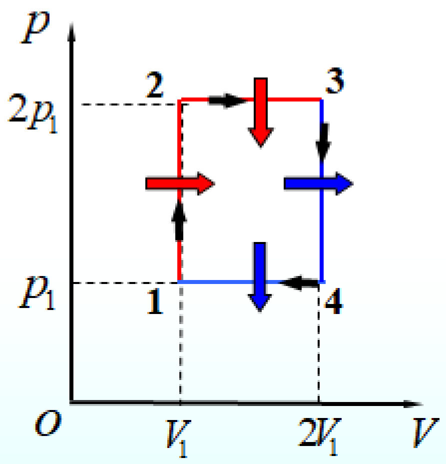
$$W = (2p_1 - p_1)(2V_1 - V_1) = p_1V_1$$

$$Q_{12} = nC_V\Delta T = n\frac{f}{2}R(T_2 - T_1)$$

$$= \frac{f}{2}(2p_1V_1 - p_1V_1) = \frac{f}{2}p_1V_1$$

$$Q_{23} = nC_p \Delta T = \frac{f+2}{2} nR(T_3 - T_2)$$

$$= \frac{f+2}{2}(2p_1 \times 2V_1 - 2p_1V_1) = (f+2)p_1V_1$$



$$e = \frac{W}{|Q|} = \frac{p_1V_1}{(f+2)p_1V_1 + \frac{f}{2}p_1V_1}$$

$$= \frac{2}{3f+4} = \frac{2}{3\times 5+4} = \frac{2}{19} = 10.6\%$$

•. 热传导过程

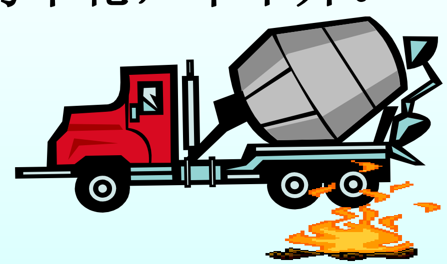
热量不能自动从低温→高温

热传导过程具有方向性。

• 功热转换过程

剎车摩擦生热。

烘烤车轮，车不开。



功热转换过程具有方向性

•气体的绝热自由膨胀过程

自由膨胀，不可自动收缩

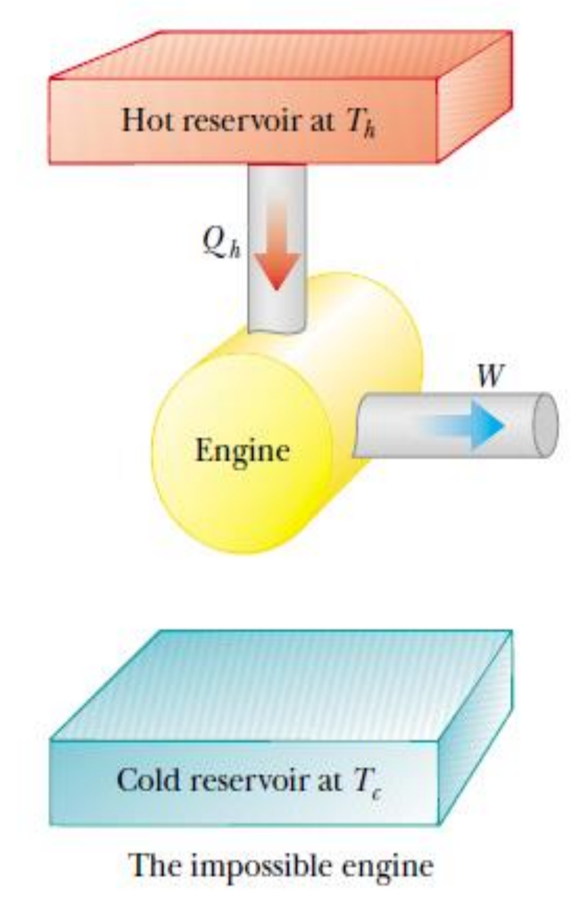
气体绝热自由膨胀的过程具有方向性

实际宏观过程按一定方向进行的规律就是热力学第二定律。

15.2 Second Law of Thermodynamics

1. Two statements

(1) The Kelvin-Plank Statement:

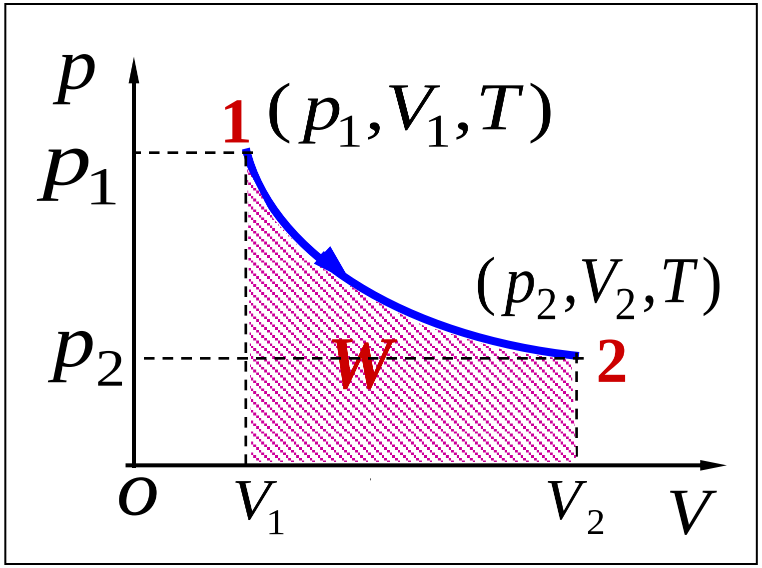


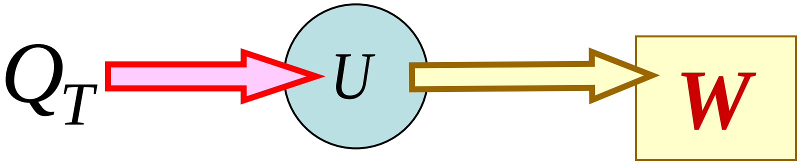
The impossible engine

No device is possible whose sole effect is to transform a given amount of heat completely into work.

There are no perfect (100%-efficient) engines .

不可能制成一种循环工作的热机，它只从单一热源吸热，全部用与对外做功，而不使外界发生任何变化 .

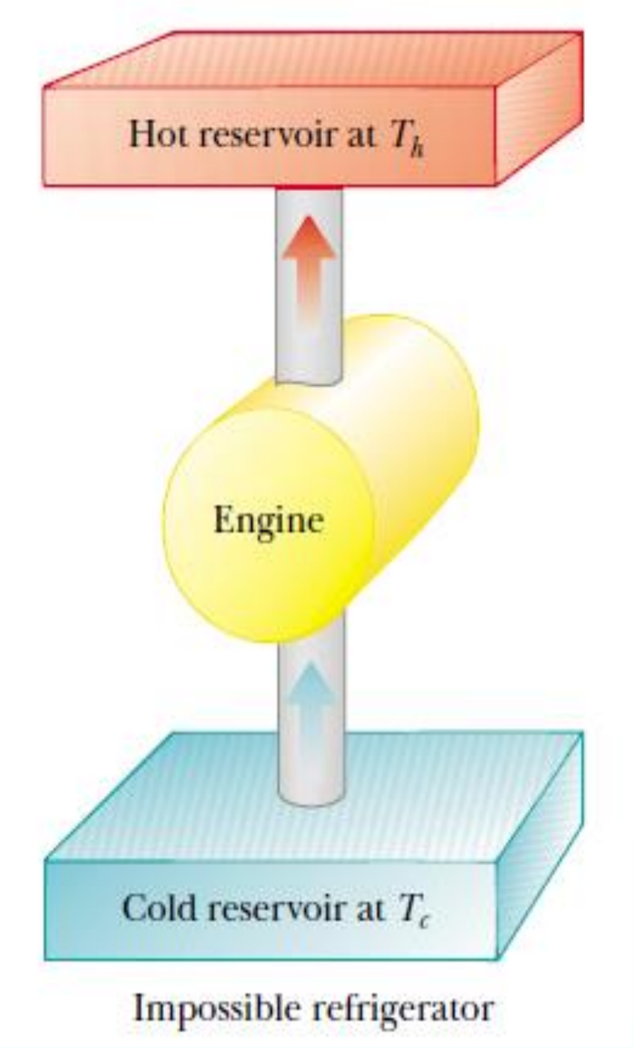




等温膨胀过程是从单一热源吸热作功，而不放出热量给其它物体，但它是非循环过程.

(2) The Clausius Statement:

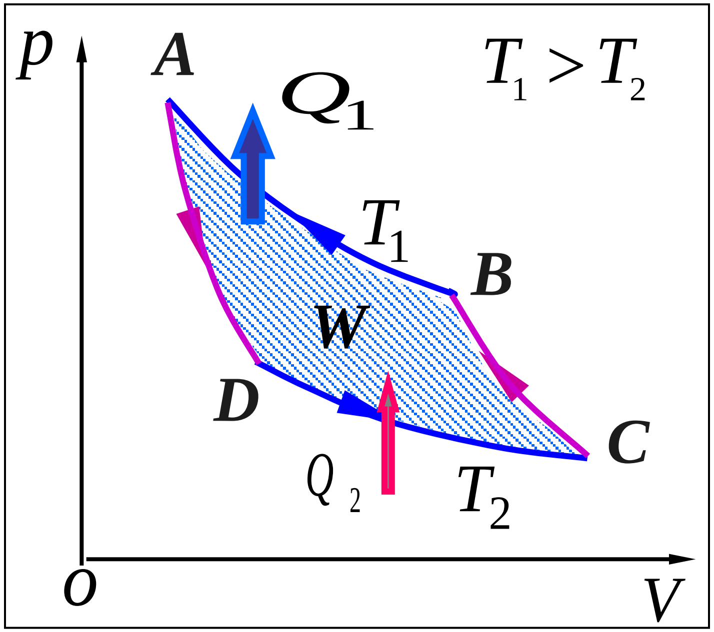
Heat will not flow **spontaneously** **from a cold object to a hot object.**

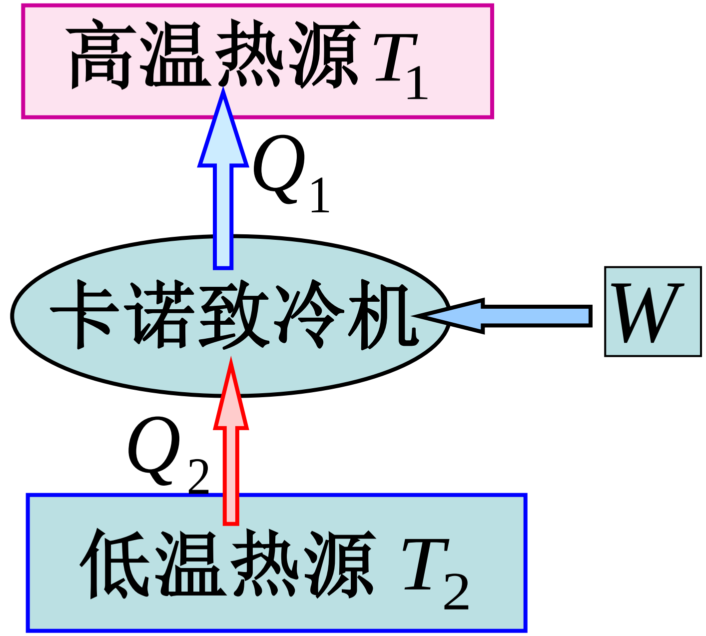


There are no perfect (workless) refrigerators.

克劳修斯表述：

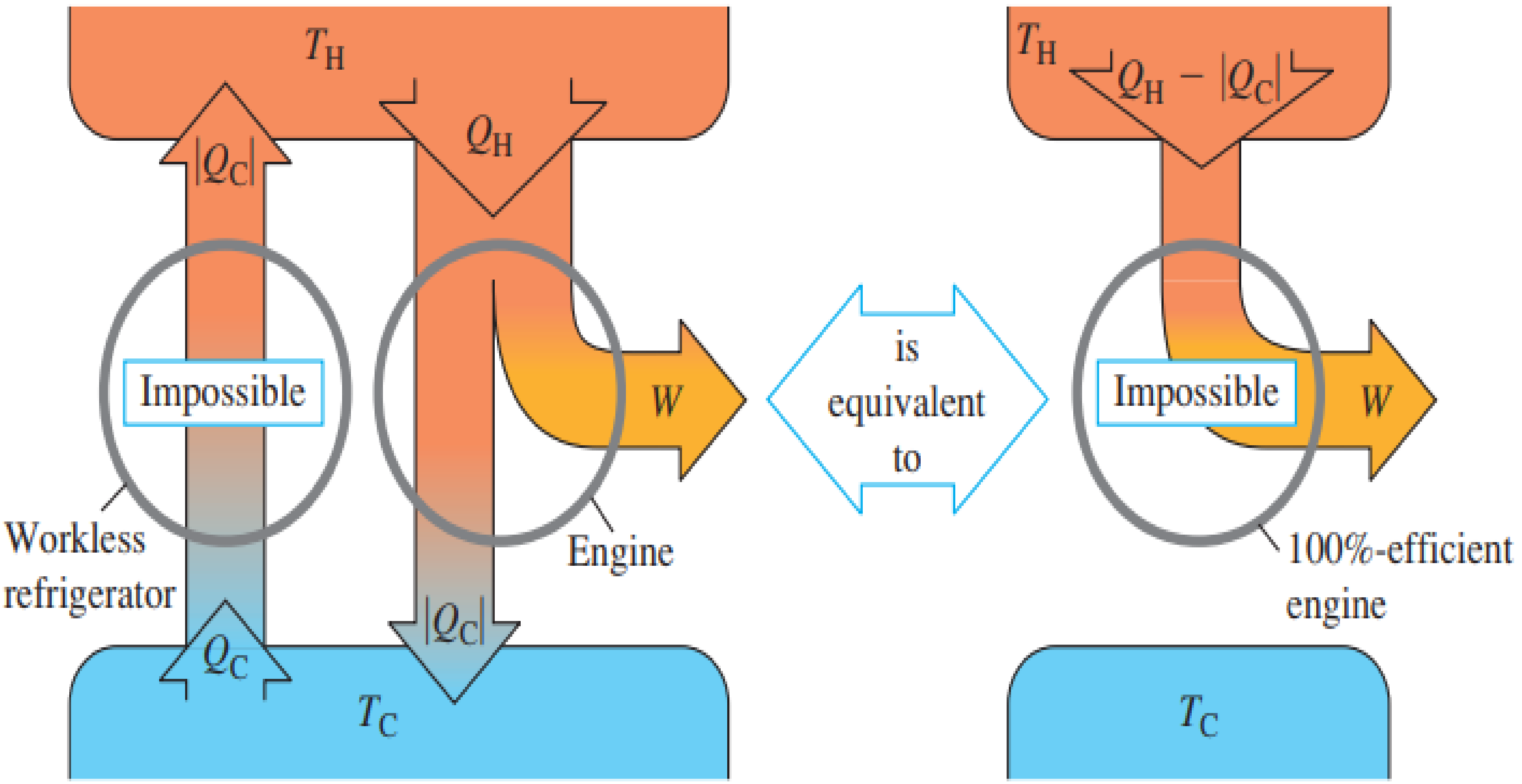
不可能把热量从低温物体自动传到高温物体而不引起外界的变化 .





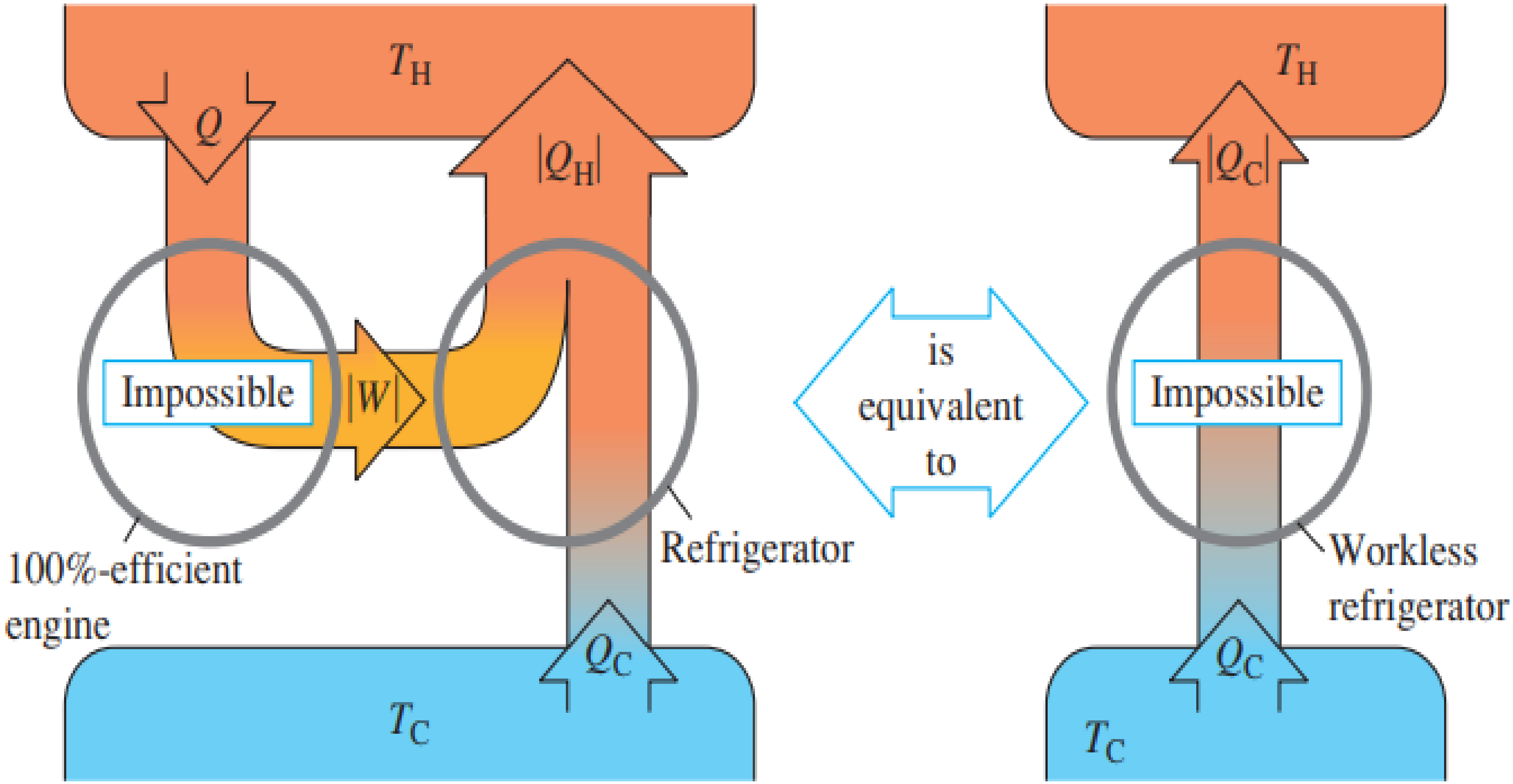
虽然卡诺致冷机能把热量从低温物体移至高温物体，但需外界作功且使环境发生变化．

(a) The "engine" statement of the second law of thermodynamics



If a workless refrigerator were possible, it could be used in conjunction with an ordinary heat engine to form a 100%-efficient engine, converting heat $Q_H - |Q_C|$ completely to work.

(b) The "refrigerator" statement of the second law of thermodynamics



If a 100%-efficient engine were possible, it could be used in conjunction with an ordinary refrigerator to form a workless refrigerator, transferring heat Q_C from the cold to the hot reservoir with no input of work.

The first law specifies that

we cannot get more energy out of a cyclic process by work than the amount of energy we put in.

"You cann't get something for nothing"

The second law states that

we cannot break even because we must put more energy in, at the higher temperature, than the net amount of energy we get out by work.

"You cann't even break even."

***Question* :** **A Carnot heat engine runs between a cold reservoir at temperature T_C and a hot reservoir at temperature T_H. You want to increase its efficiency. Of the following, which change results in the greatest increase in efficiency? The value of ΔT is the same for all changes.**

A. Raise the temperature of the hot reservoir by ΔT.
B. Raise the temperature of the cold reservoir by ΔT.
C. Lower the temperature of the hot reservoir by ΔT.
D. Lower the temperature of the cold reservoir by ΔT.
E. Raise the temperature of the hot reservoir by $\Delta T/2$. and lower the temperature of the cold reservoir by $\Delta T/2$.

Question **:A Carnot cooling device runs between a cold reservoir at temperature T_C and a hot reservoir at temperature T_H. You want to increase its coefficient of performance. Of the following, which change results in the greatest increase in coefficient? The value of ΔT is the same for all changes.**

A. Raise the temperature of the hot reservoir by ΔT.

B. Raise the temperature of the cold reservoir by ΔT.

C. Lower the temperature of the hot reservoir by ΔT.

D. Lower the temperature of the cold reservoir by ΔT.

E. Lower the temperature of the hot reservoir by $\Delta T/2$. and raise the temperature of the cold reservoir by $\Delta T/2$.

2. Reversible and Irreversible Processes;

(1) reversible process

◆ **reversible process: The system undergoing the process can be returned to its initial conditions along the same path shown on a *PV* diagram, and every point along this path is an equilibrium. It can be plotted by a continuous path on a *PV* diagram.**

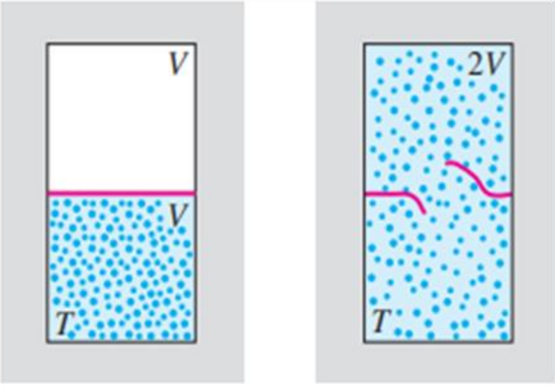
可逆过程：在系统状态变化过程中，如果逆过程能重复正过程的每一状态，而且不引起其它变化，这样的过程叫做可逆过程 .

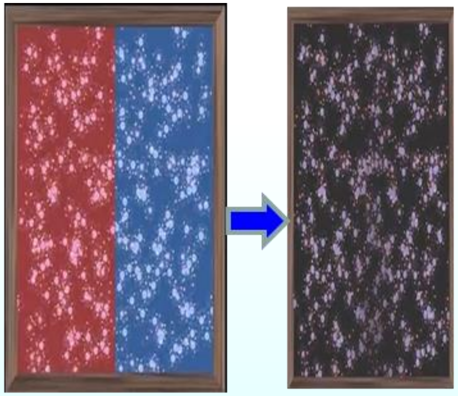
Quasistatic process +frictionless=reversible process

准静态无摩擦过程为可逆过程

◈ **Irreversible processes 不可逆过程**：如果用任何方法都不能使系统和外界恢复原来的状态，这样的过程称为不可逆过程。

Irreversible process can not be plotted by a continuous path on a *PV* diagram.

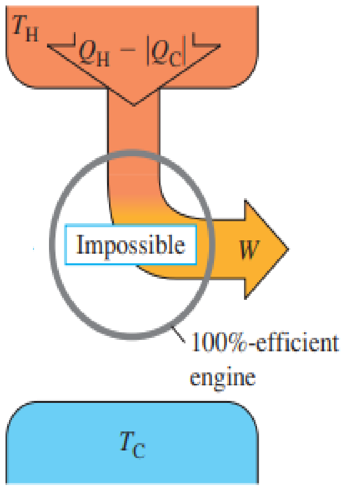




All real spontaneous process in nature are irreversible.

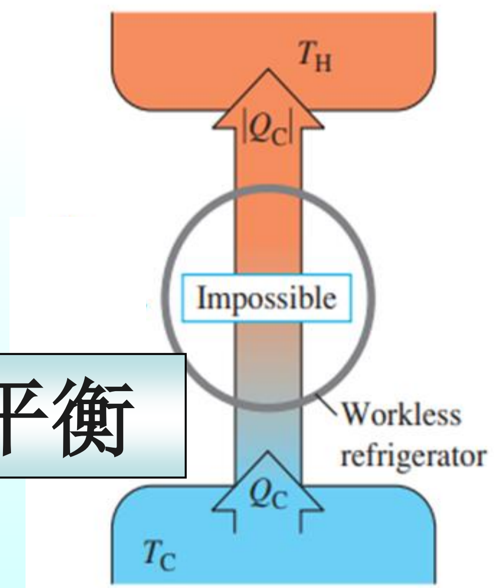
Work $\xrightarrow{\text{Completely}}$ **Heat**

有序 $\xrightarrow{\text{自发}}$ 无序



Hight *T* $\xrightarrow{\text{Spontaneously}}$ **Low *T***

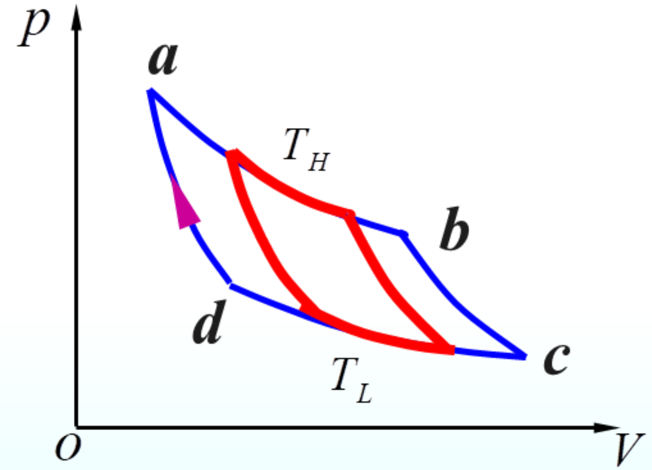
非均匀、非平衡 $\xrightarrow{\text{自发}}$ 均匀、平衡



(2) Carnot's Theorem:

All Carnot engines operating between the same two constant temperatures T_L and T_H have the same efficiency.

在相同高温热源和低温热源之间工作的任意工作物质的卡诺机都具有相同的效率.



$$e_{ideal} = 1 - \frac{T_L}{T_H}$$

The efficiency of any reversible engine operating between two heat reservoirs depends only on the temperatures of these two reservoirs, and the efficiency does not depend on the working substance.

Any irreversible engine operating between the same two fixed temperatures will have an efficiency less than this. Because they do not operate through a reversible cycle.

工作在相同的高温热源和低温热源之间的一切不可逆机的效率都不可能大于可逆机的效率 .

$$e_{real} < 1 - \frac{T_L}{T_H}$$

卡诺定理意义：

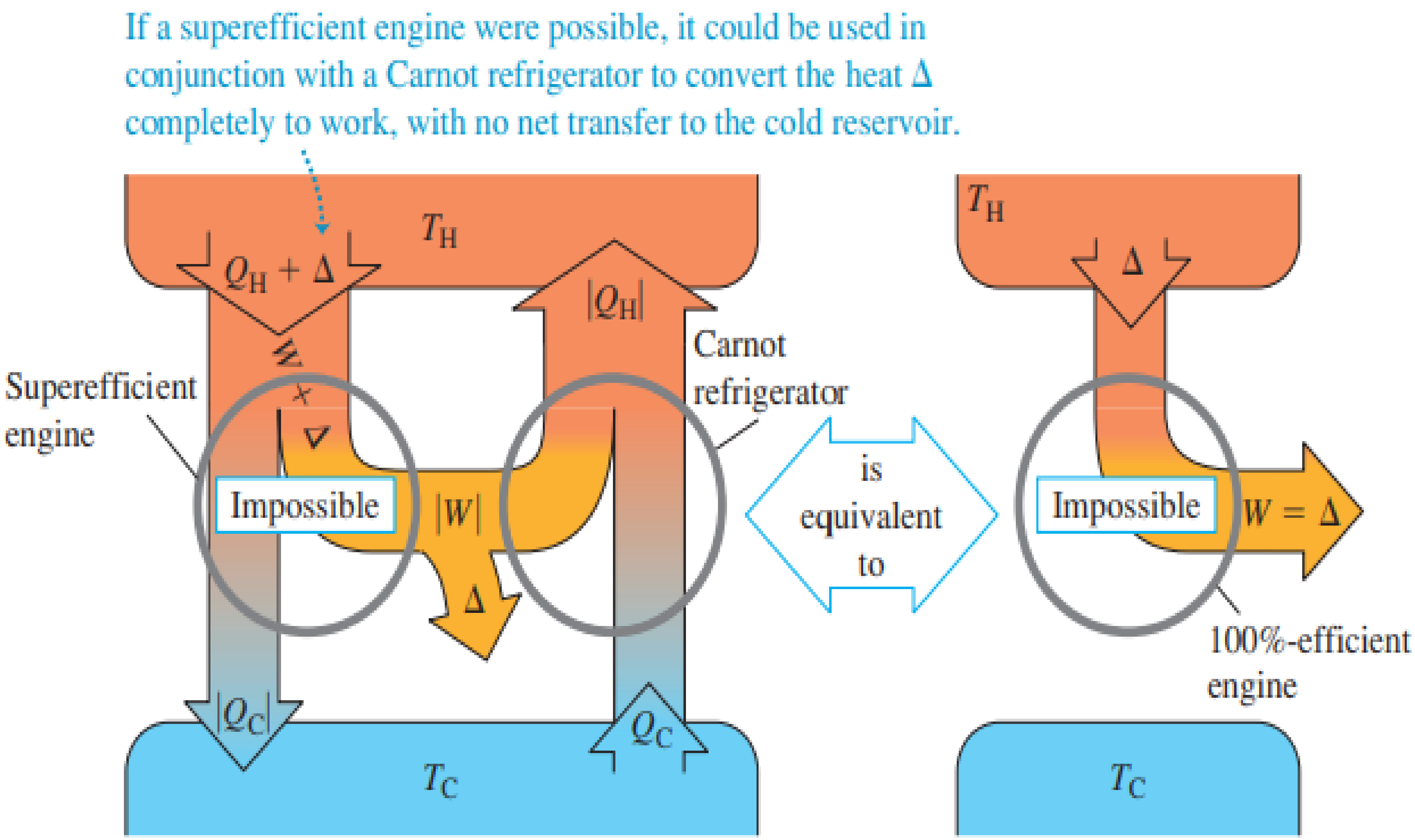
1. 给出了热机效率的极限

2. 指出提高热机效率的途径

（1）过程：尽可能接近可逆机

（2）热源：尽可能提高热源的温度差（T_L有限,提高 T_H）

Proving that the Carnot engine has the highest possible efficiency.



Question **:A Carnot heat engine and an irreversible heat engine both operate between the same high temperature and low temperature reservoirs. They absorb the same energy from the high temperature reservoir as heat. The irreversible engine:**

(A) does more work

(B) transfers more energy to the low temperature reservoir as heat

(C) has the greater efficiency

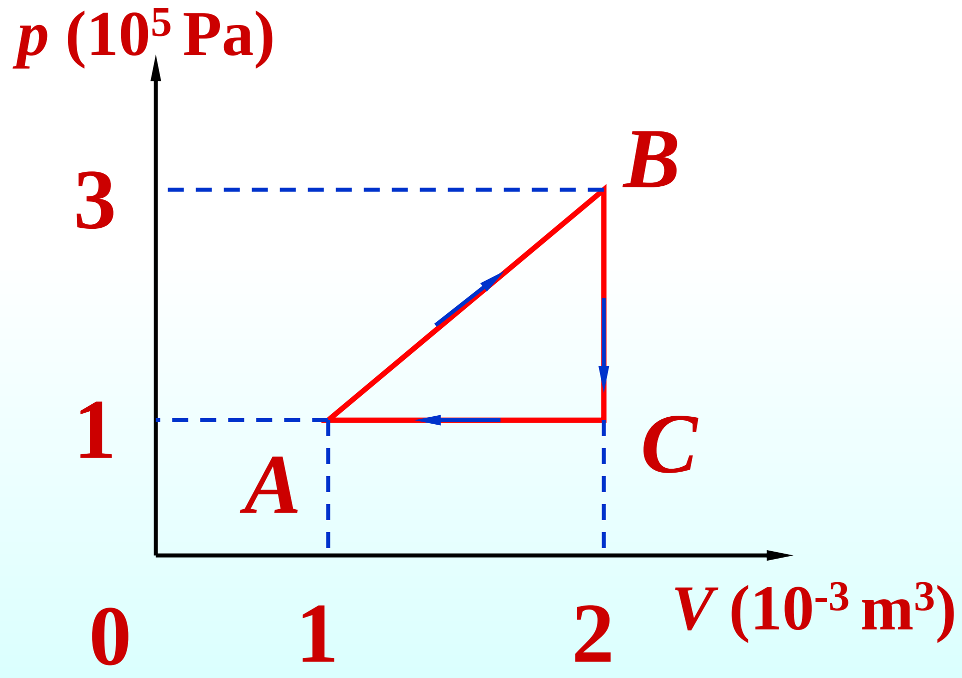
(D) has the same efficiency as the reversible engine

(E) cannot absorb the same energy from the high temperature reservoir as heat without violating the second law of thermodynamics.

Example An ideal monatomic gas, originally at point A, undergoes the following cycle as shown.

Calculate (a) the **change in internal energy**, the **work** done by the gas and the **heat added** to the gas for each process;

(b) the efficiency of this cycle.



Solution:

A—B

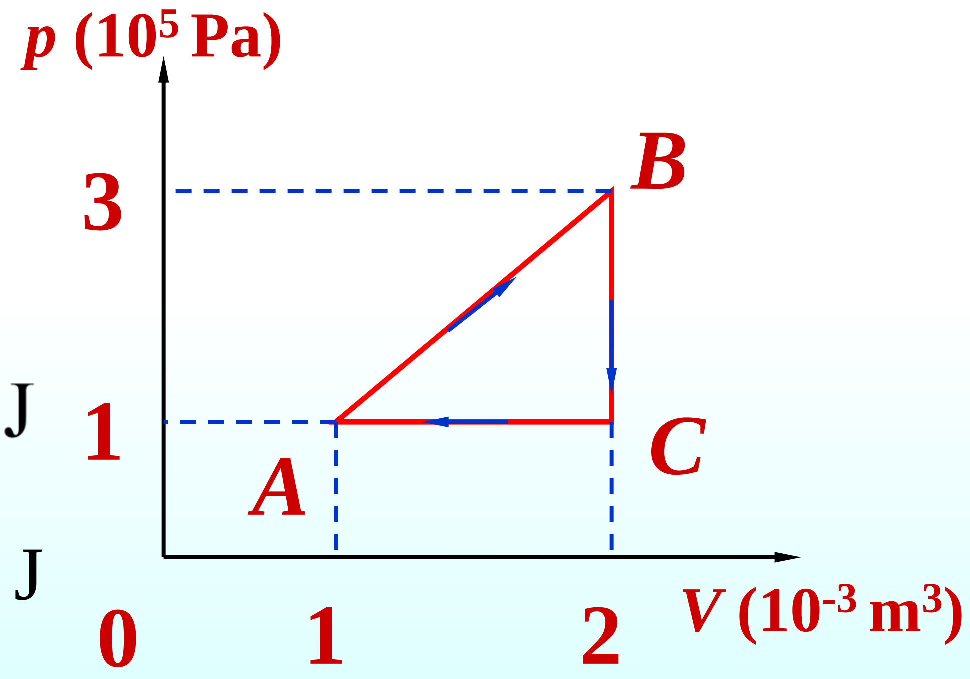
$$W_{AB}=\frac{1}{2}(p_A+p_B)(V_B-V_A)=200\text{ J}$$

$$\Delta U_{AB}=nC_V(T_B-T_A)$$

$$=n\frac{3}{2}R(T_B-T_A)$$

$$=\frac{3}{2}(p_BV_B-p_AV_A)=750\text{ J}$$

$$Q_{AB}=\Delta U_{AB}+W_{AB}=950\text{ J}$$

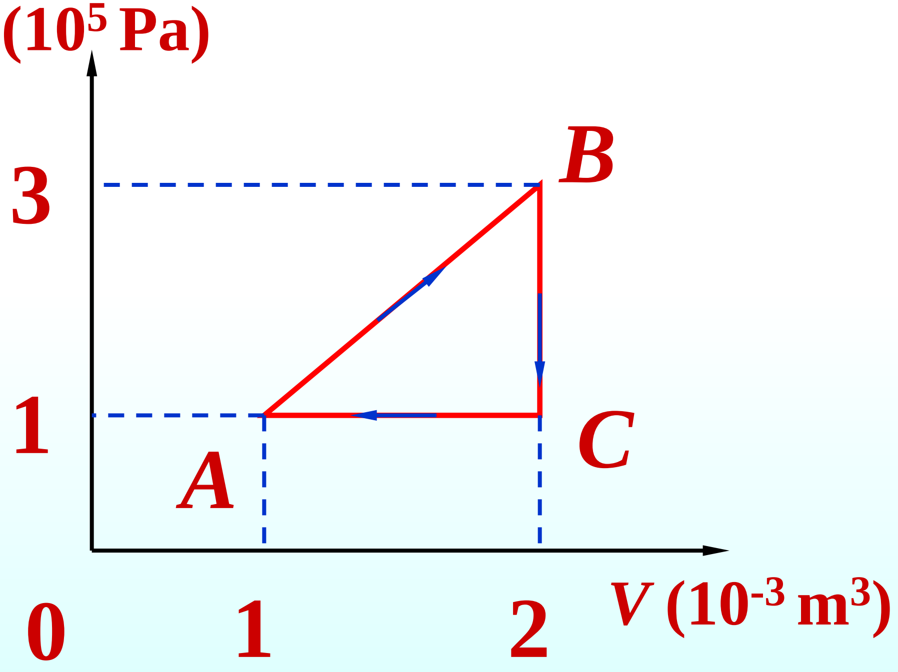


B—*C* isochoric

$$W_{BC} = 0$$

$$\Delta U_{BC} = \frac{3}{2}(p_C V_C - p_B V_B)$$

$$= -600\ \text{J}$$

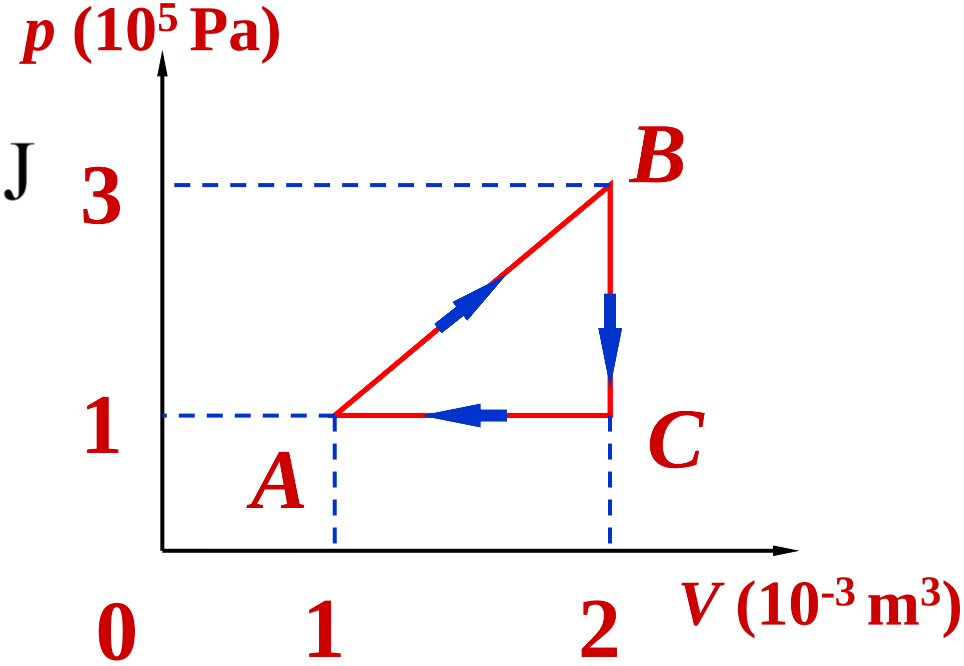


$$Q_{BC} = U_{BC} + W_{BC}$$

$$= -600\ \text{J}$$

C—A isobaric

$$W_{CA} = p_A(V_A - V_C) = -100\ \text{J}$$



$$\Delta U_{CA} = \frac{3}{2}(p_A V_A - p_C V_C)$$

$$= -150\ \text{J}$$

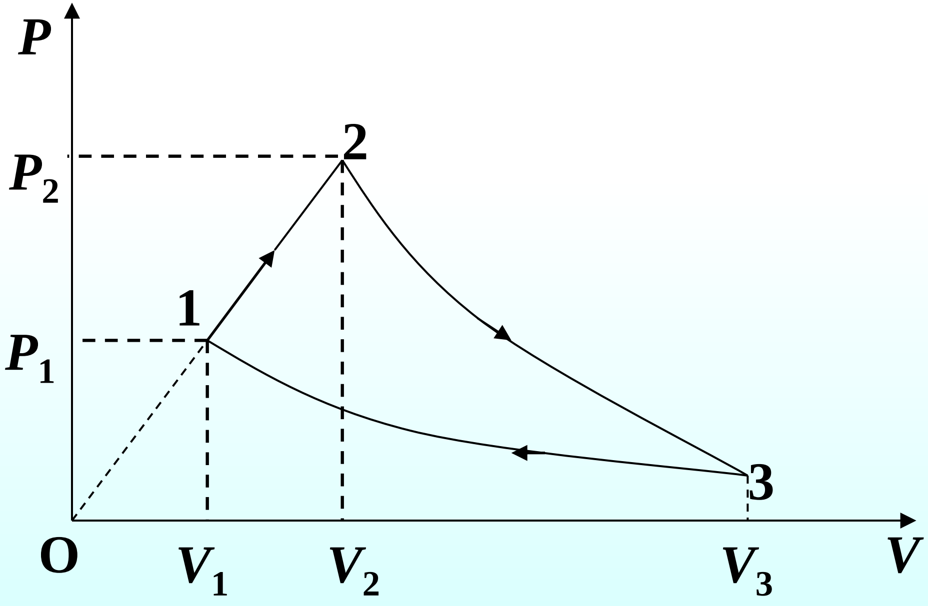
$$Q_{CA} = \Delta U_{CA} + W_{CA} = -250\ \text{J}$$

$$e = \frac{W}{Q_{\text{H}}} = \frac{W}{Q_{AB}} = \frac{100}{950} = 10.5\%$$

Two moles diatomic ideal gas takes the following cycle. In this cycle process 2-3 is adiabatic, and 3-1 is isothermal. T_1, V_1 is known and $T_2=3T_1$， $V_3=8V_1$.

Find：(a) heat in the three processes,

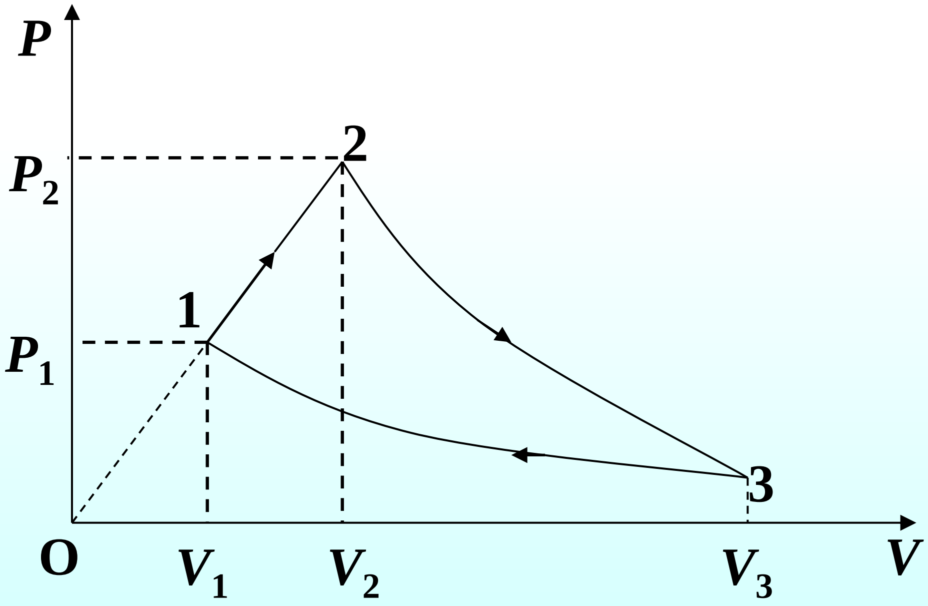
(b) the efficiency of this cycle.（ln2=0.6931）



$$1 \to 2 \qquad W_{12} = \frac{1}{2}(p_2V_2 - p_1V_1) = \frac{1}{2}(nRT_2 - nRT_1) = 2RT_1$$

$$\Delta U_{12} = \frac{f}{2}nR\Delta T_2 = \frac{f}{2}nR(3T_1 - T_1) = fnRT_1 = 10RT_1$$

$$Q_{12} = W_{12} + \Delta U_{12} = 12RT_1$$

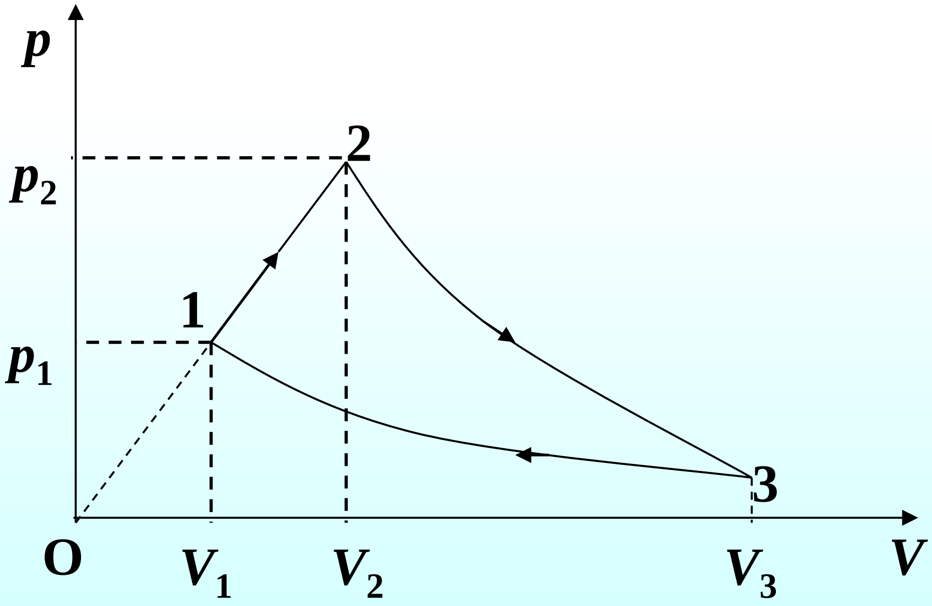


$2 \rightarrow 3$ adiabatic $Q_{23} = 0$

$3 \rightarrow 1$ isotheormal

$$Q_{31} = W_{31} + \Delta U_{31} = W_{31} = nRT_1 \ln \frac{V_1}{V_3} = -6 \ln 2 RT_1$$

$$e = 1 - \frac{|Q_2|}{Q_1} = 1 - \frac{6 \ln 2 RT_1}{12 RT_1} = 1 - \frac{\ln 2}{2} = 65.3\%$$



15.3 Entropy （熵）

1. Entropy

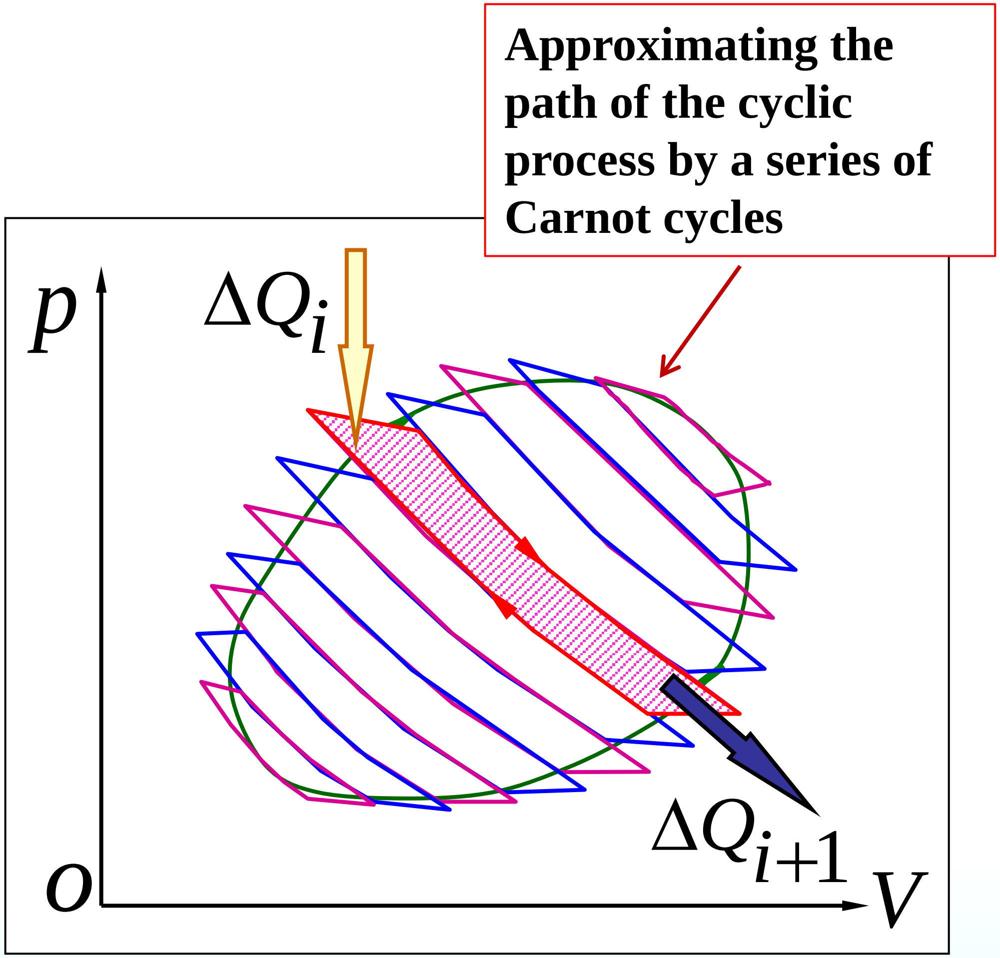
In our study of the Carnot cycle

$$\frac{|Q_H|}{T_H} = \frac{|Q_L|}{T_L}$$

Remove the absolute value signs

$$\frac{Q_H}{T_H} = -\frac{Q_L}{T_L} \Rightarrow \frac{Q_H}{T_H} + \frac{Q_L}{T_L} = 0$$

Any reversible cycle can be approximated as a series of Carnot cycles.



For each of cycles

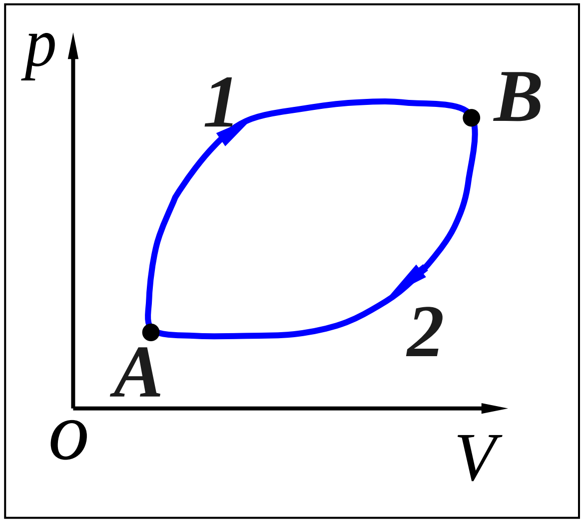
$$\frac{\Delta Q_i}{T_i} + \frac{\Delta Q_{i+1}}{T_{i+1}} = 0$$

The sum of all the cycles

$$\sum_i \frac{\Delta Q_i}{T_i} = 0$$

$$i \to \infty, \qquad \oint \frac{\mathrm{d}Q}{T} = 0$$

◆**We conclude** $\oint \frac{\mathrm{d}Q}{T} = 0$ **for any reversible cycle.**



$$\oint \frac{\mathrm{d}Q}{T} = \int_{A1B} \frac{\mathrm{d}Q}{T} + \int_{B2A} \frac{\mathrm{d}Q}{T} = 0$$

$$\int_{B2A} \frac{\mathrm{d}Q}{T} = -\int_{A1B} \frac{\mathrm{d}Q}{T}$$

$$\int_{A1B} \frac{\mathrm{d}Q}{T} = \int_{A2B} \frac{\mathrm{d}Q}{T}$$

Entropy *S* : a state variable unit：J/K

$$dS = \frac{dQ}{T}$$

Infinite reversible process

The magnitude of the entropy depends only on the state of the system, and not on the process undertaken.

The change in entropy only depends on the initial and final equilibrium states and not on the path joining them.

reversible process

$$\Delta S = S_B - S_A = \int_A^B \frac{\mathrm{d}Q}{T}$$

irreversible process

To find the entropy change for an irreversible process, replace that process with any reversible process that connects the same initial and final states.

$$\Delta S_{irre} = (S_B - S_A)_{irre} = \int_A^B \frac{\mathrm{d}Q_{re}}{T}$$

2. The principle of increase of entropy 熵增加原理

The entropy of an isolated system never decreases.

It either stays constant (reversible processes) or increases (irreversible processes).

孤立系统中的熵永不减少.

$\boxed{\Delta S \geq 0}$

irreversible processes $\Delta S > 0$

reversible processes $\Delta S = 0$

The total entropy of any system plus that of its environment increases as a result of any natural process.

$$\Delta S = \Delta S_{syst} + \Delta S_{env} > 0$$

Question

For all *reversible* processes involving a system and its environment:

(A) the entropy of the system does not change
(B) the entropy of the system increases
(C) the total entropy of the system and its environment does not change
(D) the total entropy of the system and its environment increases
(E) none of the above

Question :

For all *irreversible* processes involving a system and its environment:

(A) the entropy of the system does not change
(B) the entropy of the system increases
(C) the total entropy of the system and its environment does not change
(D) the total entropy of the system and its environment increases
(E) none of the above

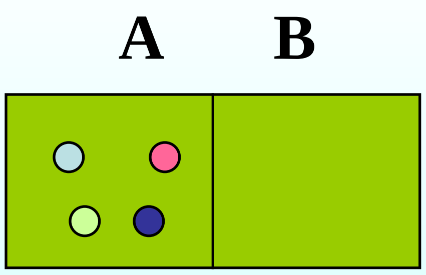
*3. Statistical interpretation of entropy and the second law

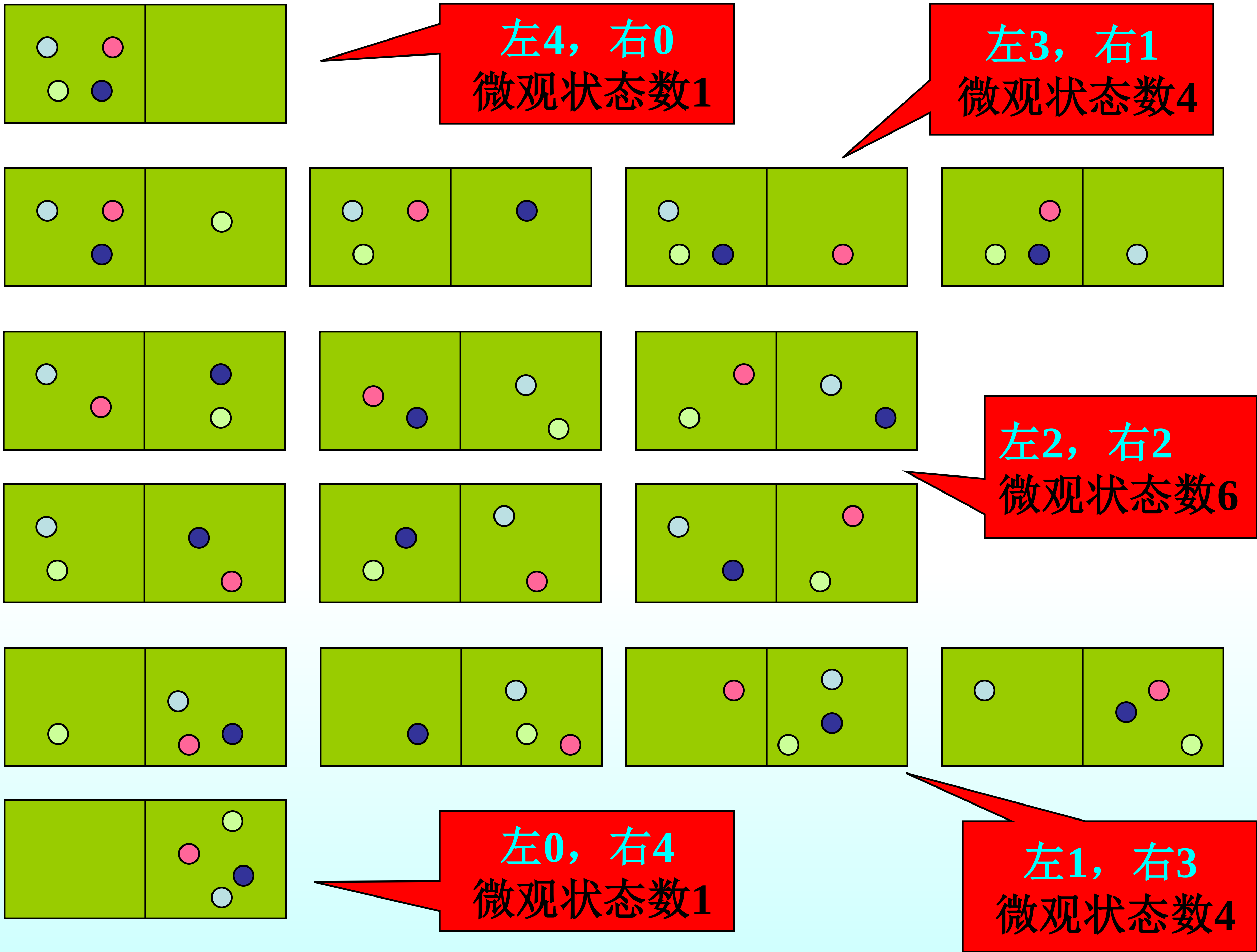
热力学第二定律的统计解释

The **microstate** of a system would be specified when the position and velocity of **every particle** is given. The **macrostate** of a system is specified by giving the **macroscopic properties**.

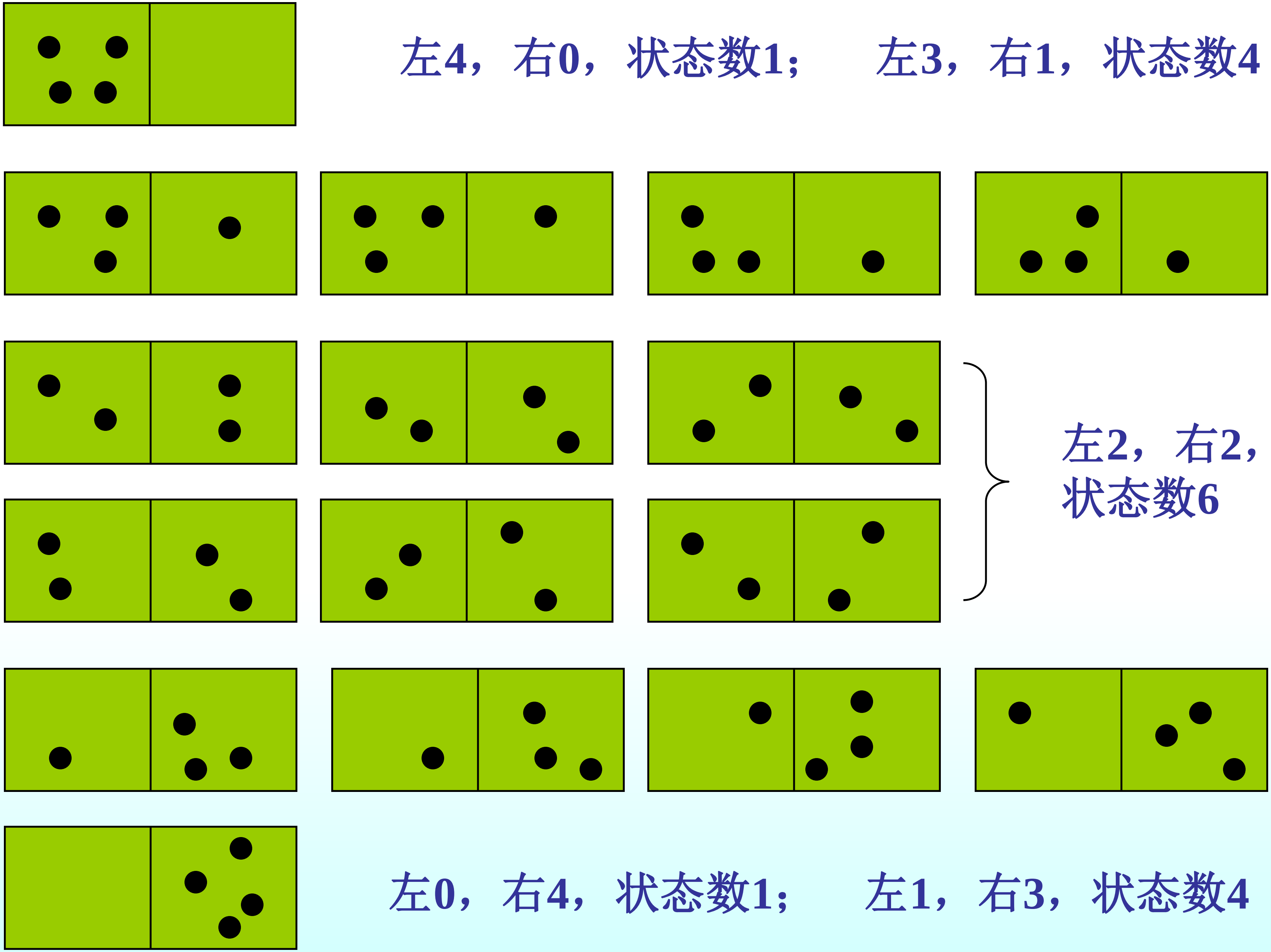
A great **many** different **microstates** can correspond to the **same macrostate**.

A basic principle behind the statistical approach is that each **microstste** is equally probable.

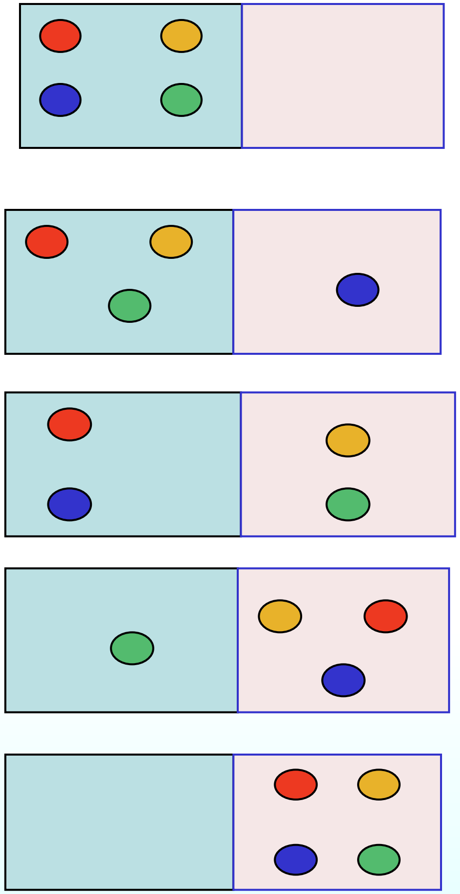




左4，右0
微观状态数1
左3，右1
微观状态数4
左2，右2
微观状态数6
左0，右4
微观状态数1
左1，右3
微观状态数4


左4，右0，状态数1；
左3，右1，状态数4
左2，右2，
状态数6
左0，右4，状态数1；
左1，右3，状态数4

$N = 4$	W	Probability
	$n_1 = 1$	$\frac{1}{16}$
	$n_2 = 4$	$\frac{4}{16}$
	$n_3 = 6$	$\frac{6}{16}$
	$n_3 = 4$	$\frac{4}{16}$
	$n_5 = 1$	$\frac{1}{16}$



Thermodynamic probability

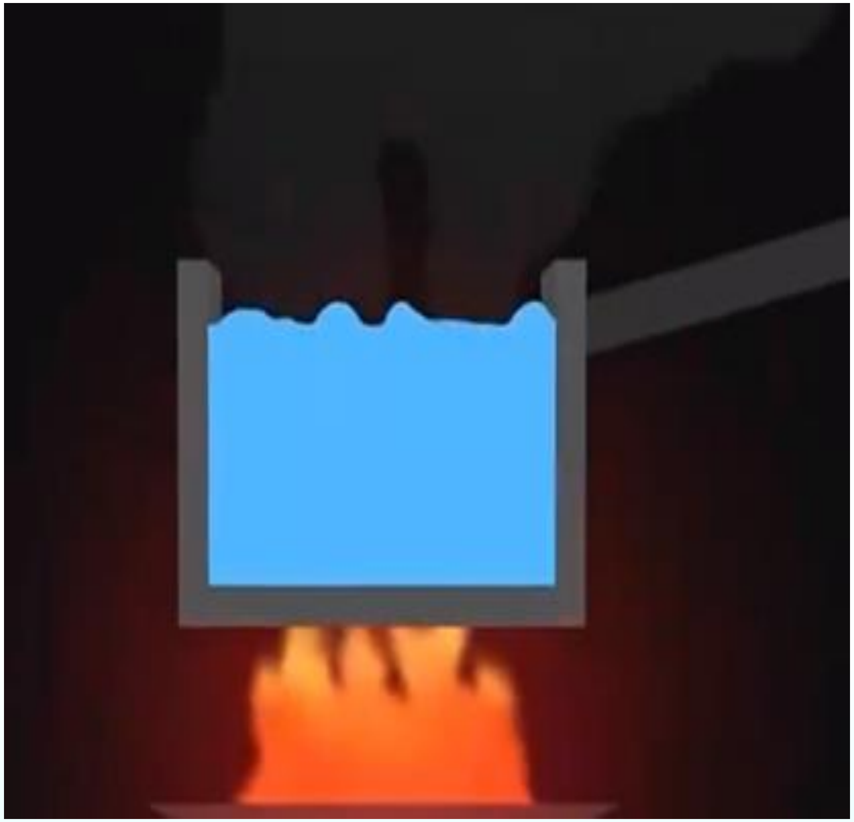
***W*: the number of microstates corresponding to the given macrostate.**

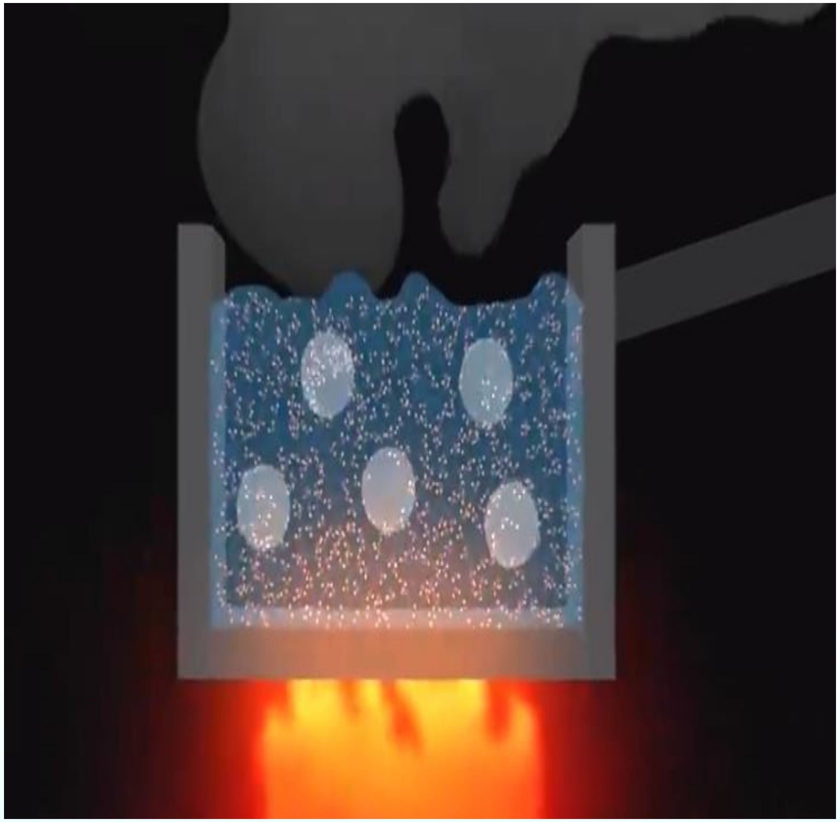
The most probable macrostate is the one with greatest *W* (disorder).

$$S = k \ln W$$

Entropy, which is a state variable, is a quantitative measure of the disorder of a system.

More about entropy





When we boil water, the energy we provide it with heat allows us to increase its entropy.

The molecules have more freedom and are in a more random distorderd behavior.

Entropy Changes in a Living Organism

When a kitten or other growing animal eats, it takes ordered chemical energy from the food and uses it to make new cells that are even more highly ordered. This process alone lowers entropy. But most of the energy in the food is either excreted in the animal's feces or used to generate heat, processes that lead to a large increase in entropy. So while the entropy of the animal alone decreases, the total entropy of animal plus food increases.

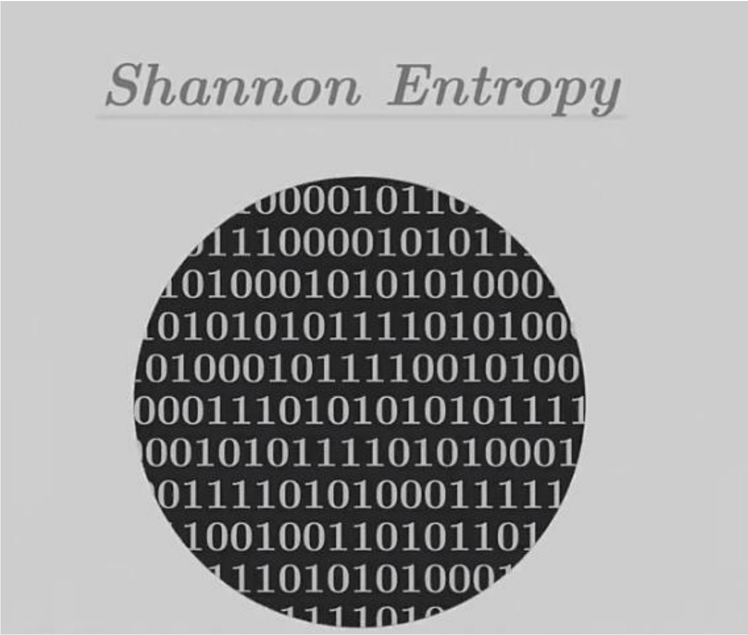
Entropy and disorder

Natural processes tend to move toward a state of **greater disorder**.

Entropy and energy availability

An increase in entropy means a loss of the ability to do work. The universal usefulness of energy decreases—**energy is degraded**.

Shannon Entropy (Information Entropy)



Shannon's entropy, measures the amount of information contained in an object such as text.

abc.abc.abc.abc.abc.abc.abc.
abc.abc.abc.abc.abc.abc.abc.
abc.abc.abc.abc.abc.abc.abc.
abc.abc.abc.abc.

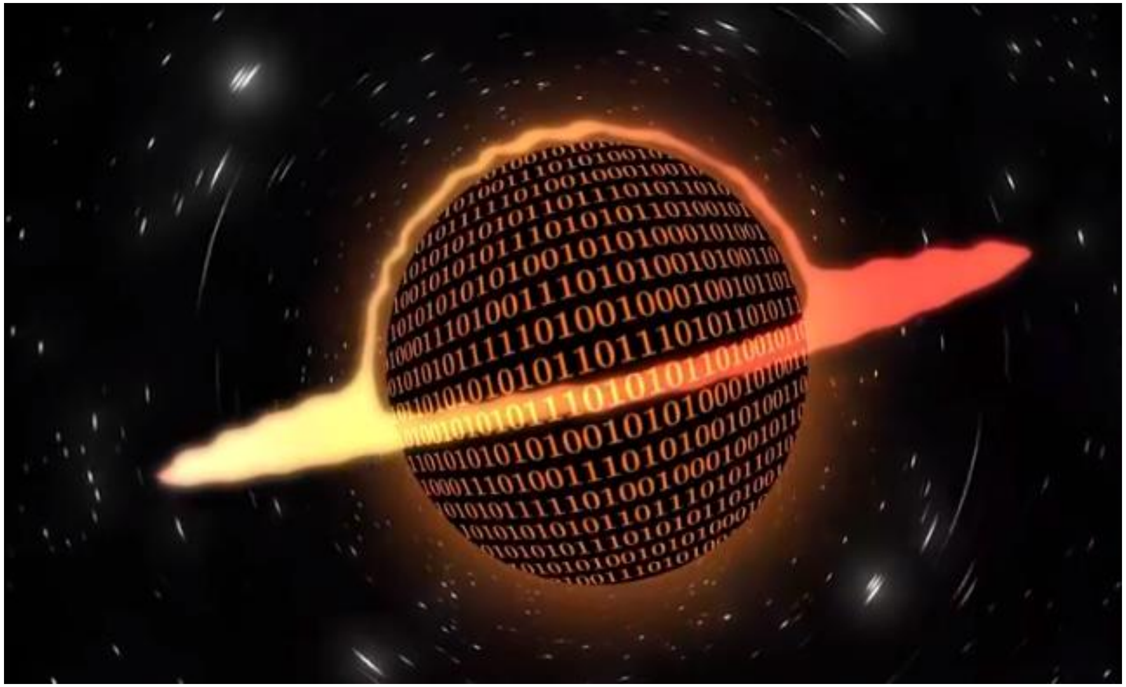
A repeating text contains little information.

tfnciaPgkghaz,cbHfjupocfea.
yfnrfiflv,;cfp/odlpmd 5125hc
uitj hry4riag6 7cIaBCcu795/
yuabgixpmr.%cag

A more random less structured text contains a lot of information.

Information is negentropy.

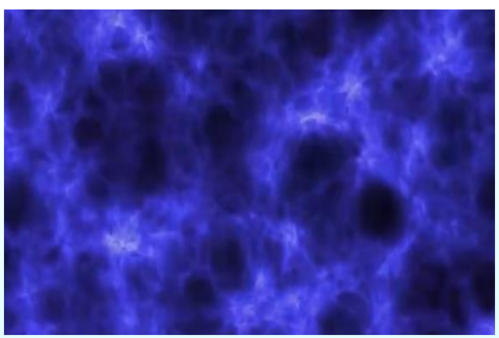
Entropy of a black hole



The entropy of a black hole is distributed over its surface, and the more information it captures by absorbing new objects, the larger the balck hole becomes.

Entropy –the arrow of time

When we let a system evolve spontaneously, its entropy tends to increase, making the system more and more homegeneous.



Heat death

As time goes by, the structures in our universe have tendency to blur and the univese will approch a state of maximum disorder. All change will cease.

Question

According to the second law of thermodynamics:

(A) heat energy cannot be completely converted to work

(B) work cannot be completely converted to heat energy

(C) for all cyclic processes we have dQ/T < 0

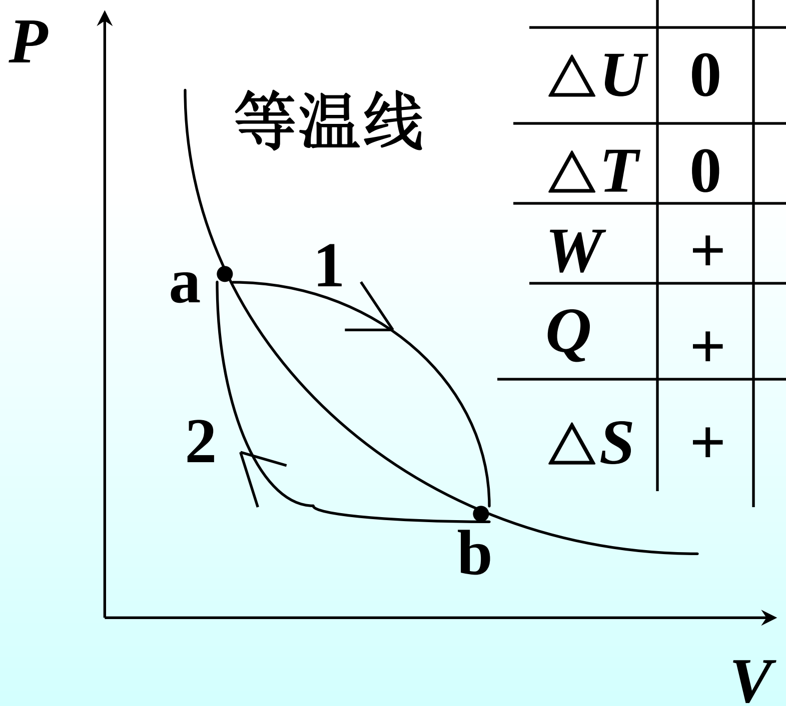
(D) the reason all heat engine efficiencies are less than 100% is friction, which is unavoidable

(E) all of the above are true

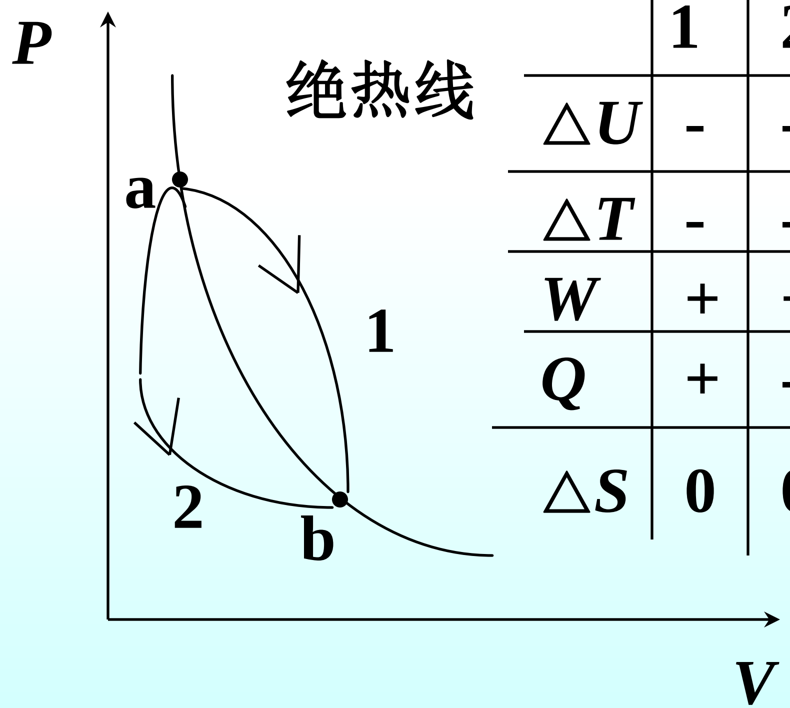
***Question* :** **A engine manufacturer makes the following claims: the heat input per second of the engine is 9.0kJ at 475K. The heat output per second is 4.0kJ at 325K. Do you believe these claims?**

***Question* :** **Two identical systems are taken from state *a* to state *b* by two different irreversible processes. Will the change in entropy for the system be the same for each process?**

例：理想气体经历下述过程，讨论$\triangle U$，$\triangle T$，$\triangle S$，W 和 Q 的符号。



	1	2
$\triangle U$	0	0
$\triangle T$	0	0
W	+	-
Q	+	-
$\triangle S$	+	-

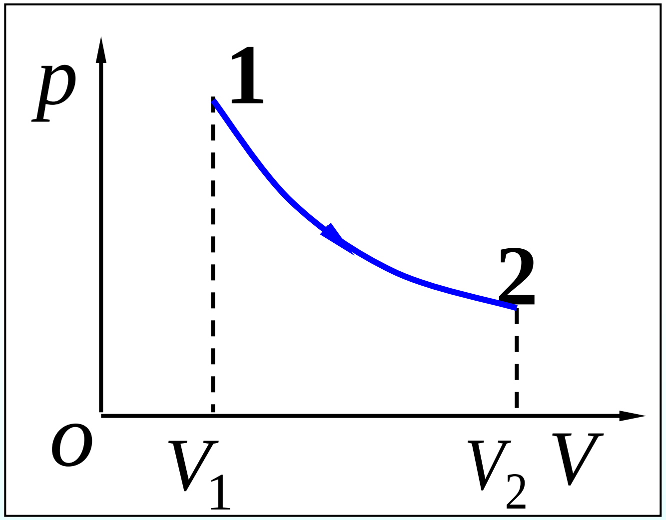


	1	2
$\triangle U$	-	-
$\triangle T$	-	-
W	+	+
Q	+	-
$\triangle S$	0	0

Entropy change in a free expansion

Consider the adiabatic free expansion of *n* moles of an idea gas from V_1 to volume V_2, where $V_2 > V_1$.

Calculate the change in entropy (a) of the gas and (b) of the surrounding environment.



$$(a)\Delta S_{gas} = S_2 - S_1$$

$$= \int_1^2 \frac{\mathrm{d}Q}{T} = \int_1^2 \frac{p\mathrm{d}V}{T} = \int_{V_1}^{V_2} nR\frac{\mathrm{d}V}{V}$$

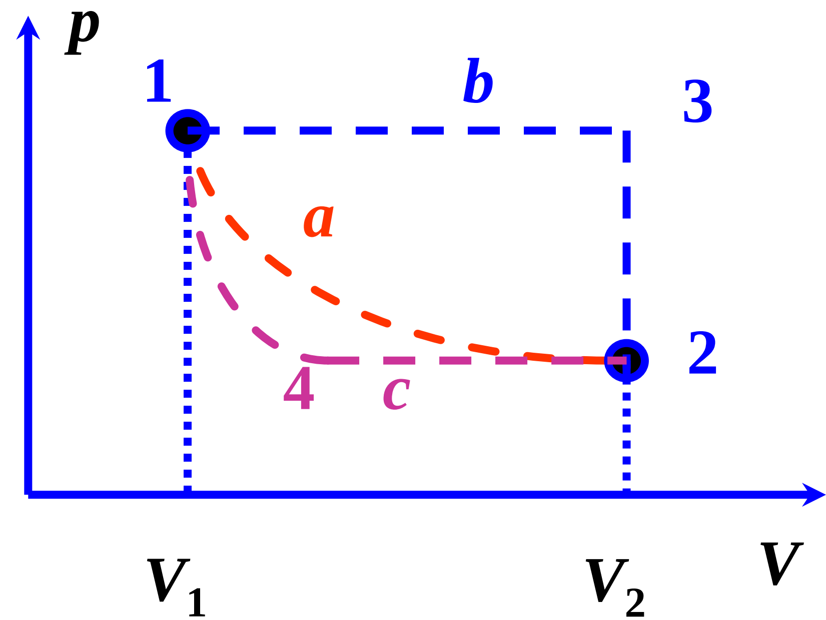
$$= nR\ln\frac{V_2}{V_1} > 0$$

irreversible

Think of a reversible isothermal process from volume V_1 to V_2

$$(b)\Delta S_{env} = 0$$

Another way 1-3 isobaric，3-2 isochoric



$$\Delta S=\int\frac{\mathrm{d}Q}{T}=n\int_1^3\frac{C_P\,\mathrm{d}T}{T}+n\int_3^2\frac{C_V\,\mathrm{d}T}{T}$$

$$=nC_V\int_1^3\frac{\mathrm{d}T}{T}+nR\int_1^3\frac{\mathrm{d}T}{T}+nC_V\int_3^2\frac{\mathrm{d}T}{T}$$

$$=nC_V\left(\ln T_3-\ln T_1+\ln T_2-\ln T_3\right)+nR\ln\frac{T_3}{T_1}\qquad(\because T_1=T_2)$$

$$=nR\ln\frac{T_3}{T_1}=nR\ln\frac{V_2}{V_1}$$

1→3, isobaric:

$V_1/T_1=V_3/T_3=V_2/T_3$

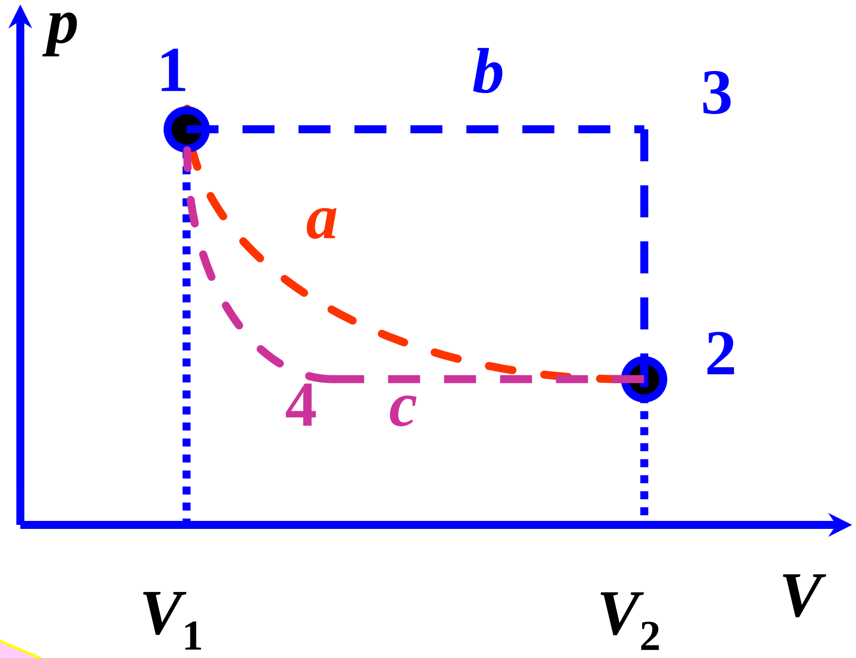
1-4 adiabatic， 4-2 isobaric

$$\Delta S = 0 + \int_4^2 \frac{\mathrm{d}Q}{T}$$

$$= \int_4^2 \frac{nC_P\,\mathrm{d}T}{T} = nC_P \ln\frac{T_2}{T_4}$$

$$= n\frac{\gamma\ \ R}{\gamma - 1}\ln\frac{T_2}{T_1}\left(\frac{P_1}{P_4}\right)^{\frac{\gamma-1}{\gamma}}$$

$$= nR\ln\frac{P_1}{P_2} = nR\ln\frac{V_2}{V_1}$$



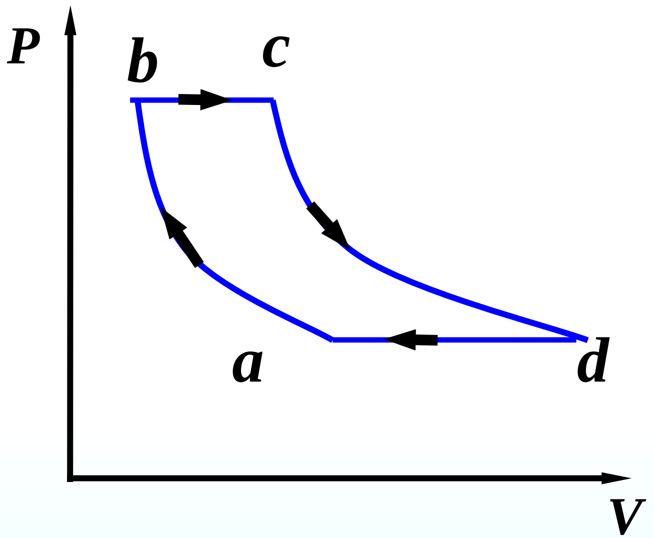
1→2, isothermal:

$p_1V_1=p_2V_2$

1→4, adiabatic:

$$p_4^{\gamma-1}T_4^{-\gamma} = p_1^{\gamma-1}T_1^{-\gamma}$$

Question : **A heat engine takes cycle as shown in the Fig. Paths ab and cd are isothermal(等温的); bc and ad are isobaric(等压的) . Which one is not right?(不定项选择题）**



(A) $\left|W_{bc}\right| = \left|W_{da}\right|$

(B) $\left|Q_{bc}\right| = \left|Q_{da}\right|$

(C) $\left|\Delta U_{bc}\right| = \left|\Delta U_{da}\right|$

(D) $\left|Q_{cd}\right| > \left|Q_{ab}\right|$

(E) $\left|W_{cd}\right| = \left|W_{ab}\right|$

Question **: Four Carnot engines（卡诺热机） that operates between the temperatures of the following choices, which has grestest efficiency（效率）?**

(A) 200K and 400K

(B) 250K and 450K

(C) 300K and 400K

(D) 400K and 600K

(E) 500K and 700K

Question: How many of the following statements are right?

It is possible for heat to flow even if the internal energies of the two objects are the same.

It is possible for the temperature of a system to remain constant even though heat flows into or out of it?

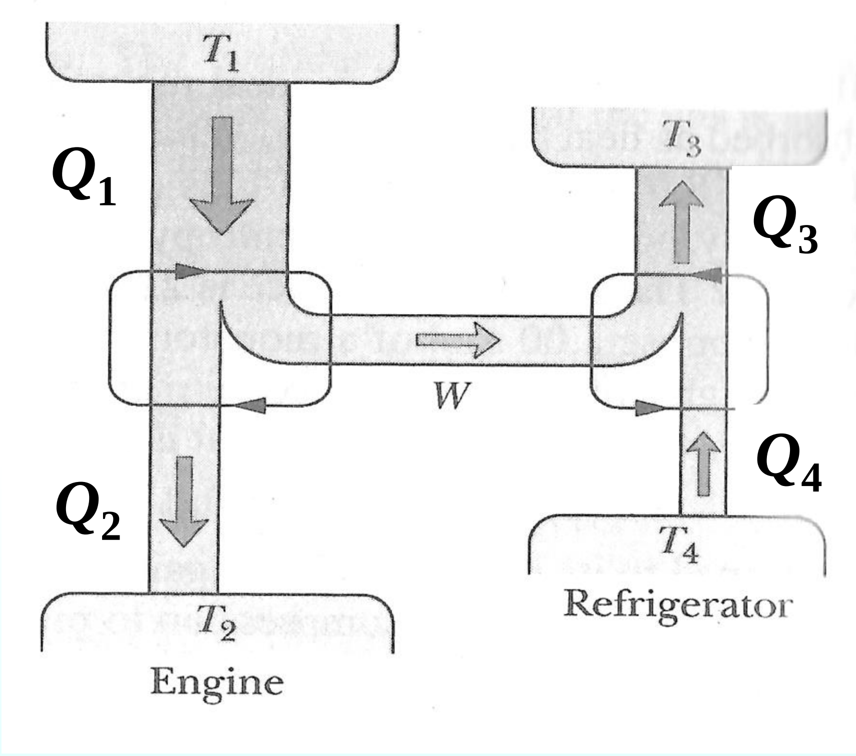
In an reversible（可逆的） isothermal process, 1000J of work is done by an ideal gas. This is enough information to tell how much heat has been added to the system.

For free expansion(自由膨胀) in which an ideal gas is allowed to expand in volume adiabatically(绝热的) , the work done by system zero .

(A) 0　　(B) 1　　(C) 2　　(D) 3　　(E) 4

Example: Figure represents a Carnot engine that works between temperatures T_1=400K and T_2=150K and drives a Carnot refrigerator that works between temperatures between T_3=325K and T_4=225K. What is the ratio Q_3/Q_1?

$$e = \frac{W}{|Q_1|} = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1}$$



$$CP = \frac{|Q_4|}{|W|} = \frac{|Q_3| - |W|}{|W|} = \frac{T_4}{T_3 - T_4}$$

$$\frac{|Q_3|}{|W|} = \frac{T_3}{T_3 - T_4}$$

$$\frac{|Q_3|}{|Q_1|} = \frac{T_1 - T_2}{T_1}\frac{T_3}{T_3 - T_4} = \frac{400 - 150}{400}\frac{325}{325 - 225} = \frac{65}{32}(2.03)$$

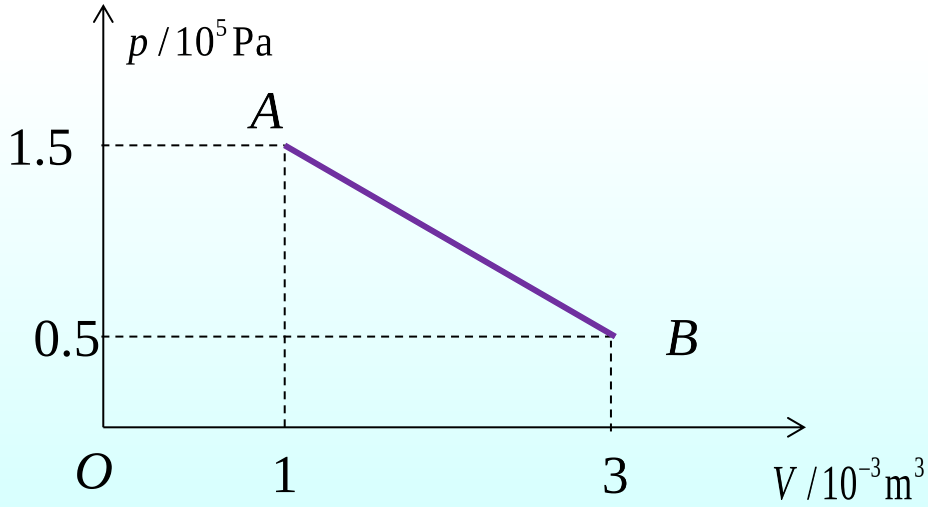
Example:

0.1mol的单原子理想气体，经历如图所示过程，由状态A经一直线到达状态B，

(1)在此过程中，气体吸热多少？

(2)在此过程中，哪一点温度最高？

(3) 在此过程中经历每一微小变化时，气体是否总是吸热？吸热放热的转折点在哪里？



$$T_A = T_B \quad \Delta U_{AB} = Q_{AB} - W_{AB} = 0$$

(1) $Q = W = 200\text{J}$

(2) $p = -0.5V + 2$ $$T = \frac{pV}{nR} = \frac{V(-0.5V+2)}{nR}$$

$$\frac{dT}{dV} = 0 \rightarrow V = 2\times10^{-3}\,\text{m}^3$$ ***T*** **取最高温度**

(3) $$dQ = dU + pdV = \frac{f}{2}nRdT + pdV = 0$$

$$\frac{f}{2}(pdV + Vdp) + pdV = 0 \rightarrow (\frac{f}{2}+1)\,pdV + \frac{f}{2}Vdp = 0$$

$$(\frac{\mathrm{d}p}{\mathrm{d}V})_A = -\gamma\frac{p_A}{V_A} = -0.5 \Rightarrow$$

$$-\frac{5}{3}\frac{(-0.5V+2)}{V} = -0.5 \Rightarrow V = 2.5\times10^{-3}\,\text{m}^3$$